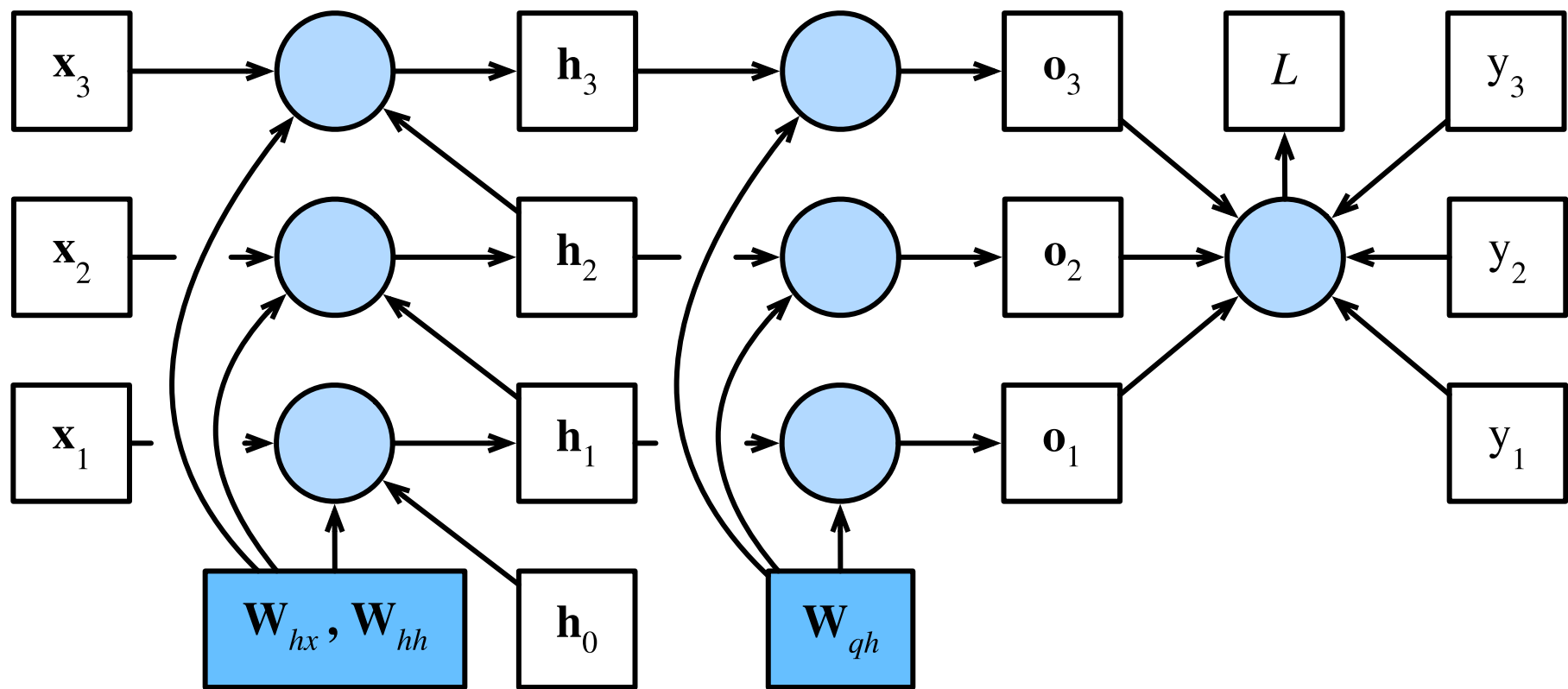
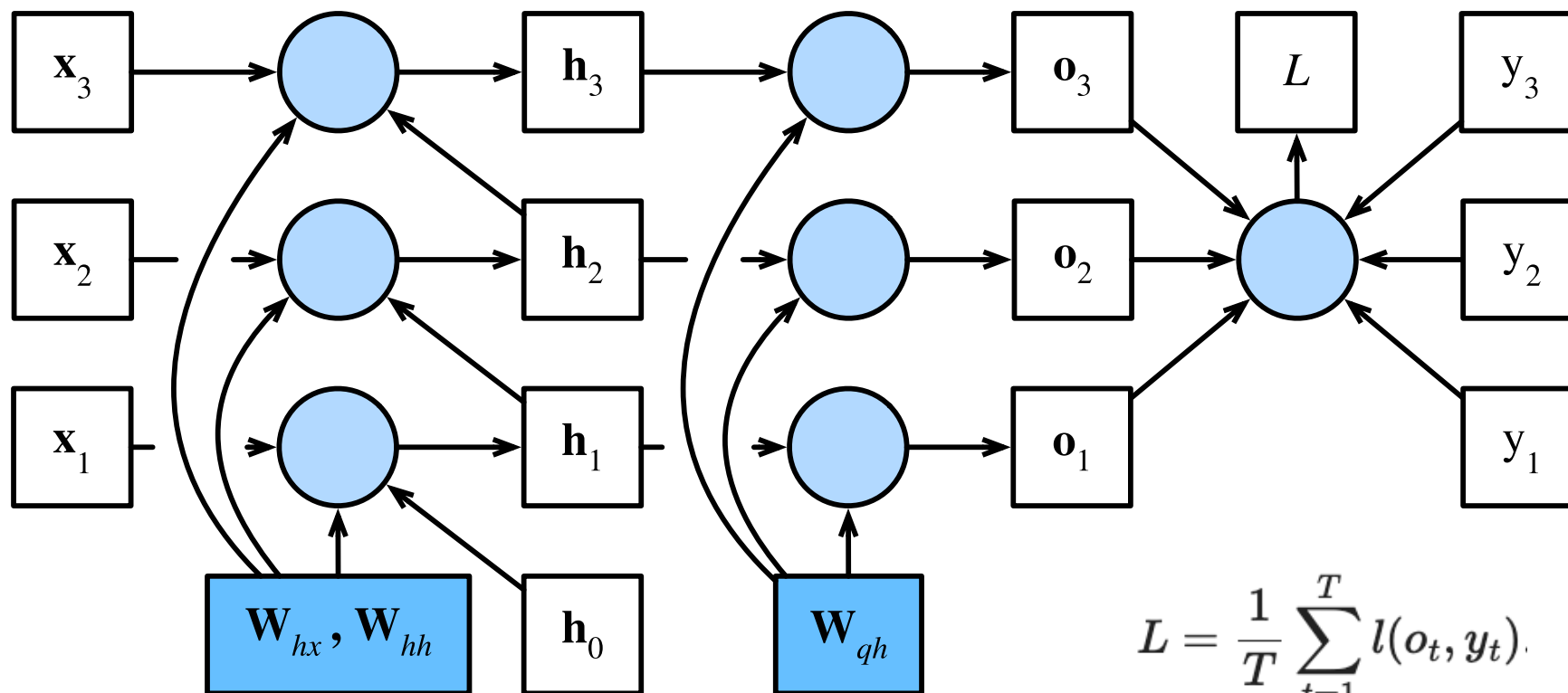


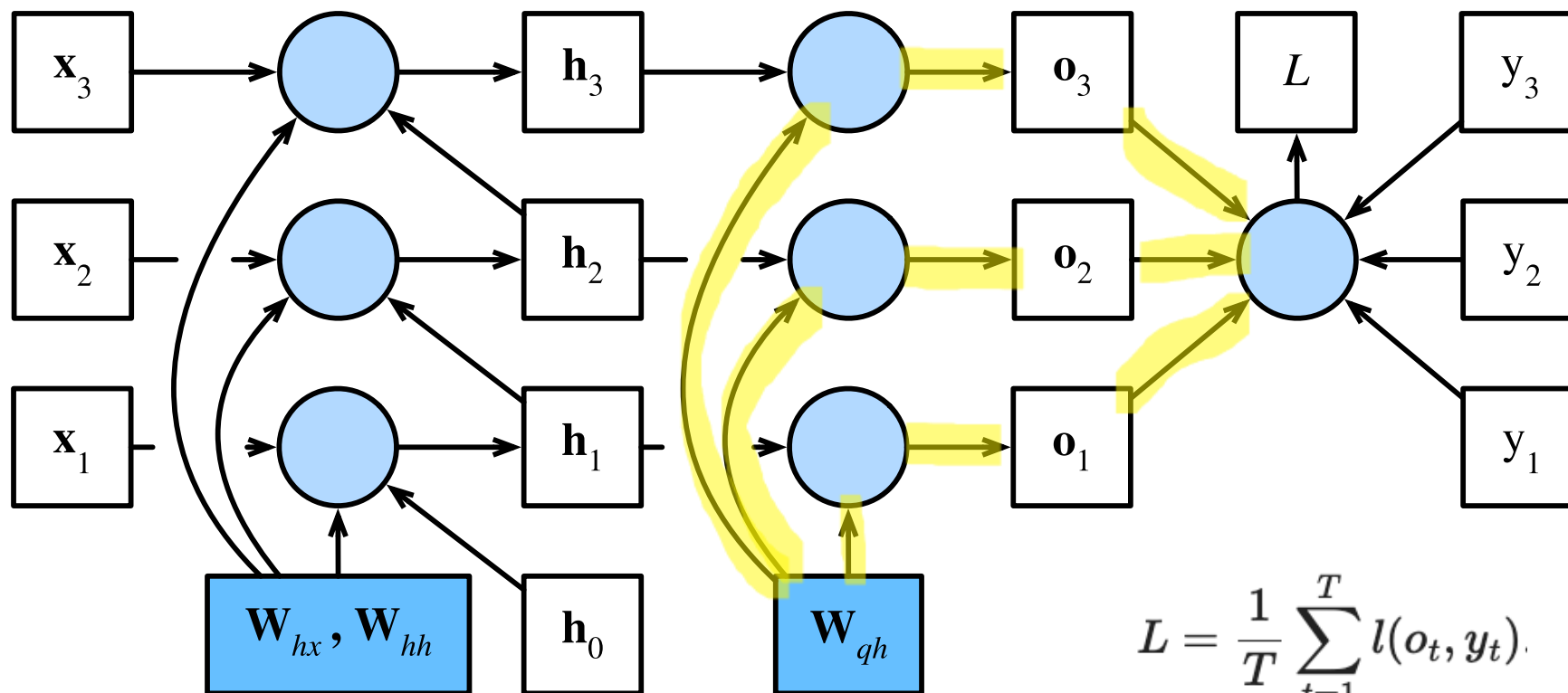


BPTT



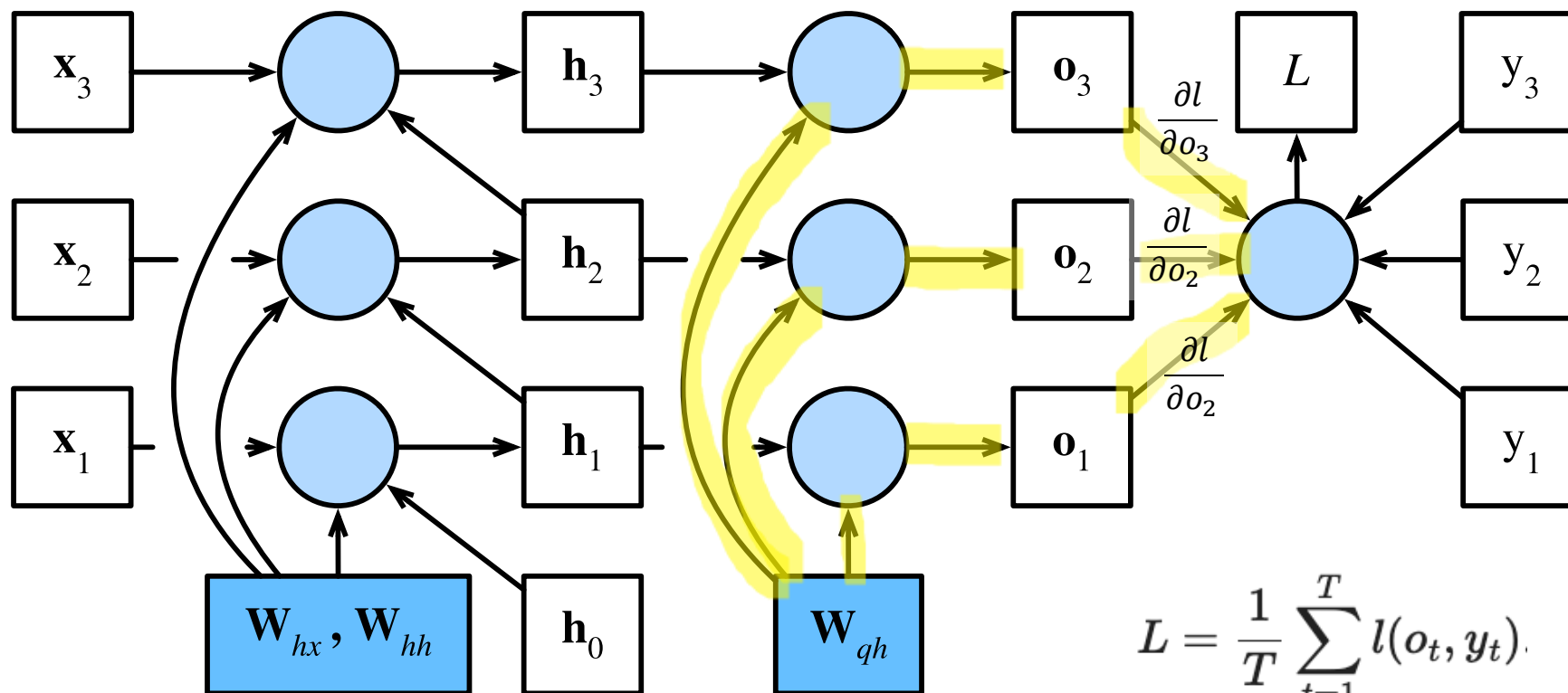
Step 1: $\frac{\partial L}{\partial \mathbf{W}_{qh}}$

BPTT



Step 1: $\frac{\partial L}{\partial \mathbf{W}_{qh}}$

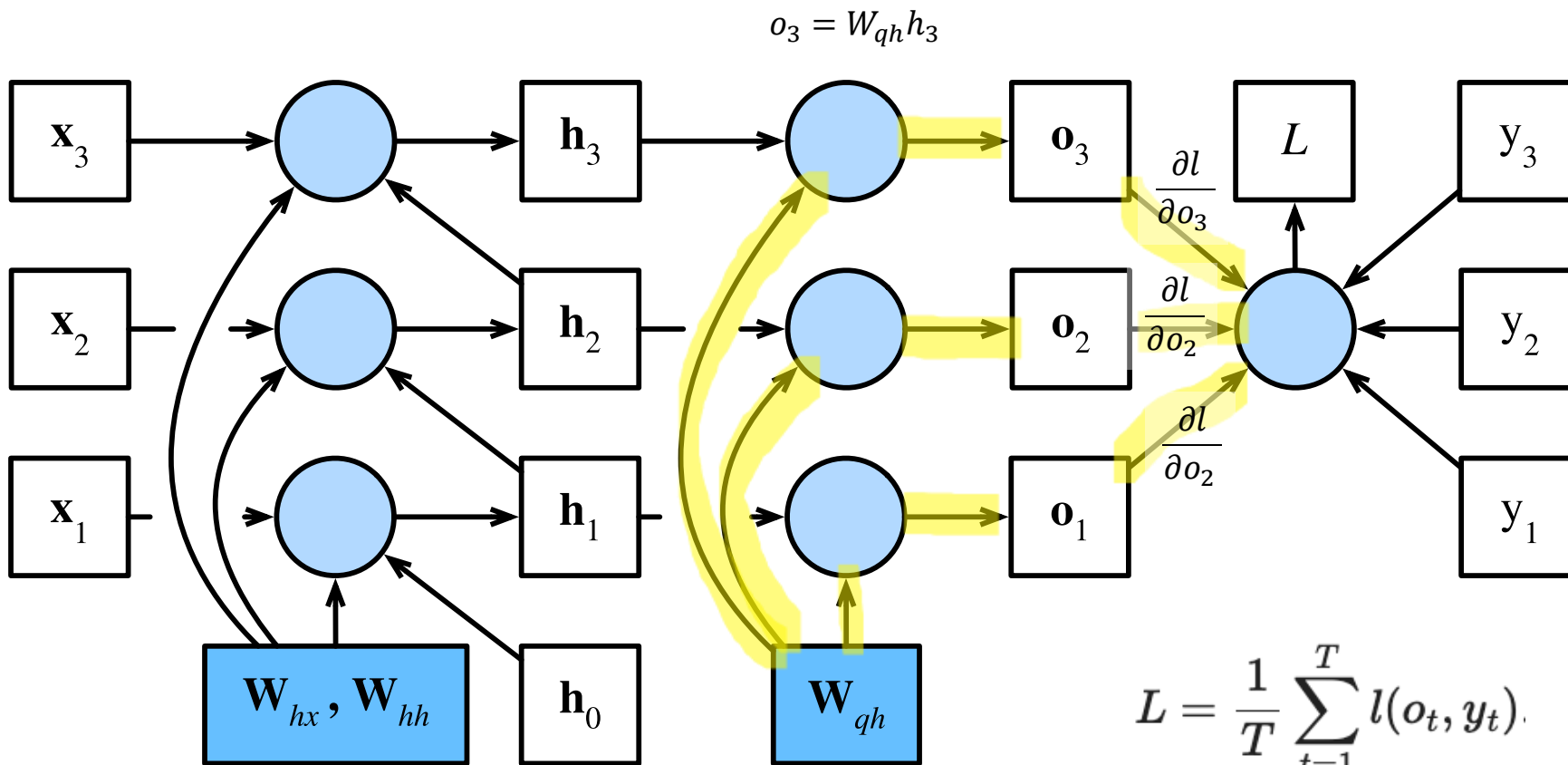
BPTT



$$L = \frac{1}{T} \sum_{t=1}^T l(o_t, y_t).$$

Step 1: $\frac{\partial L}{\partial W_{qh}}$

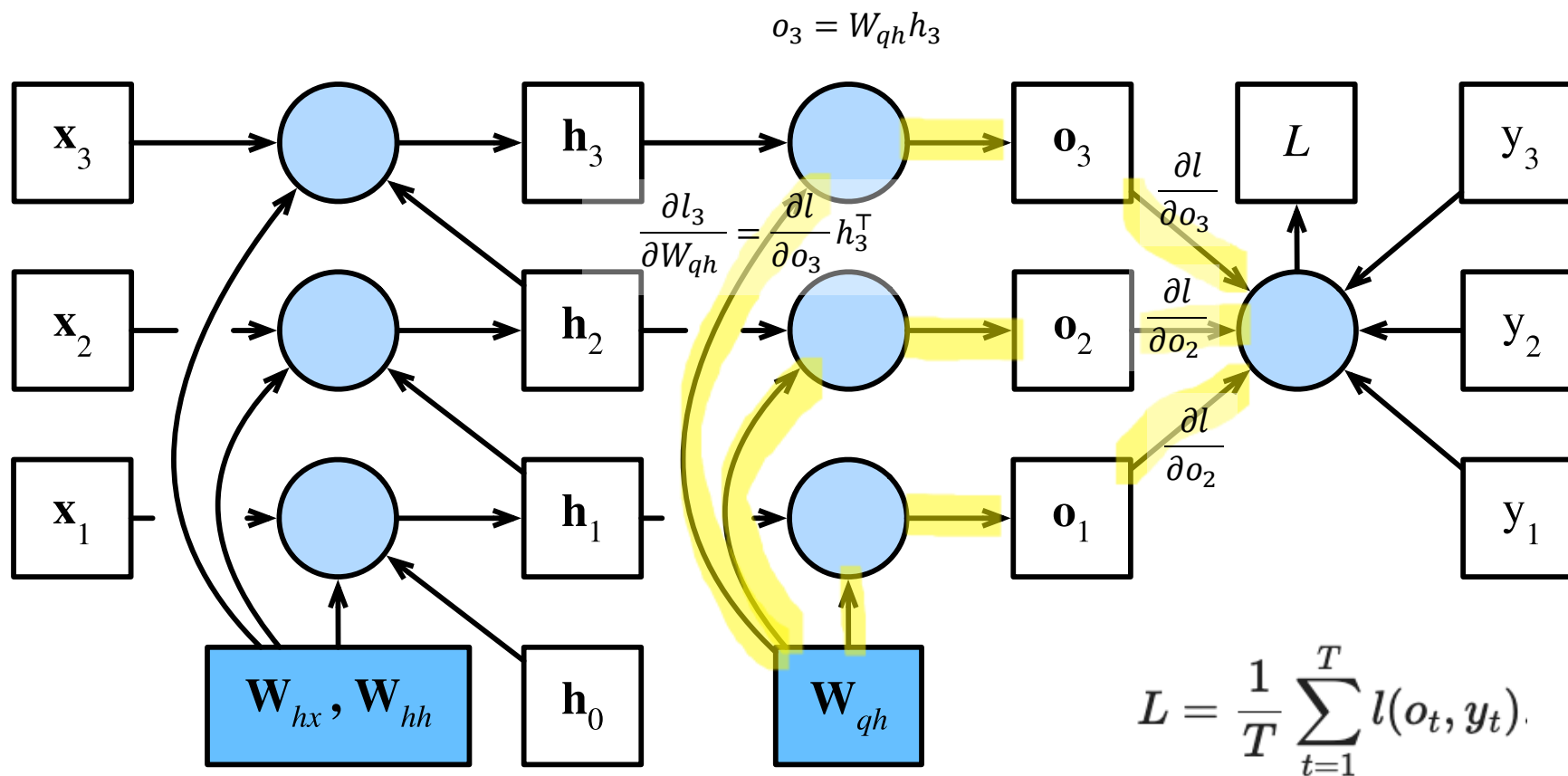
BPTT



$$L = \frac{1}{T} \sum_{t=1}^T l(o_t, y_t).$$

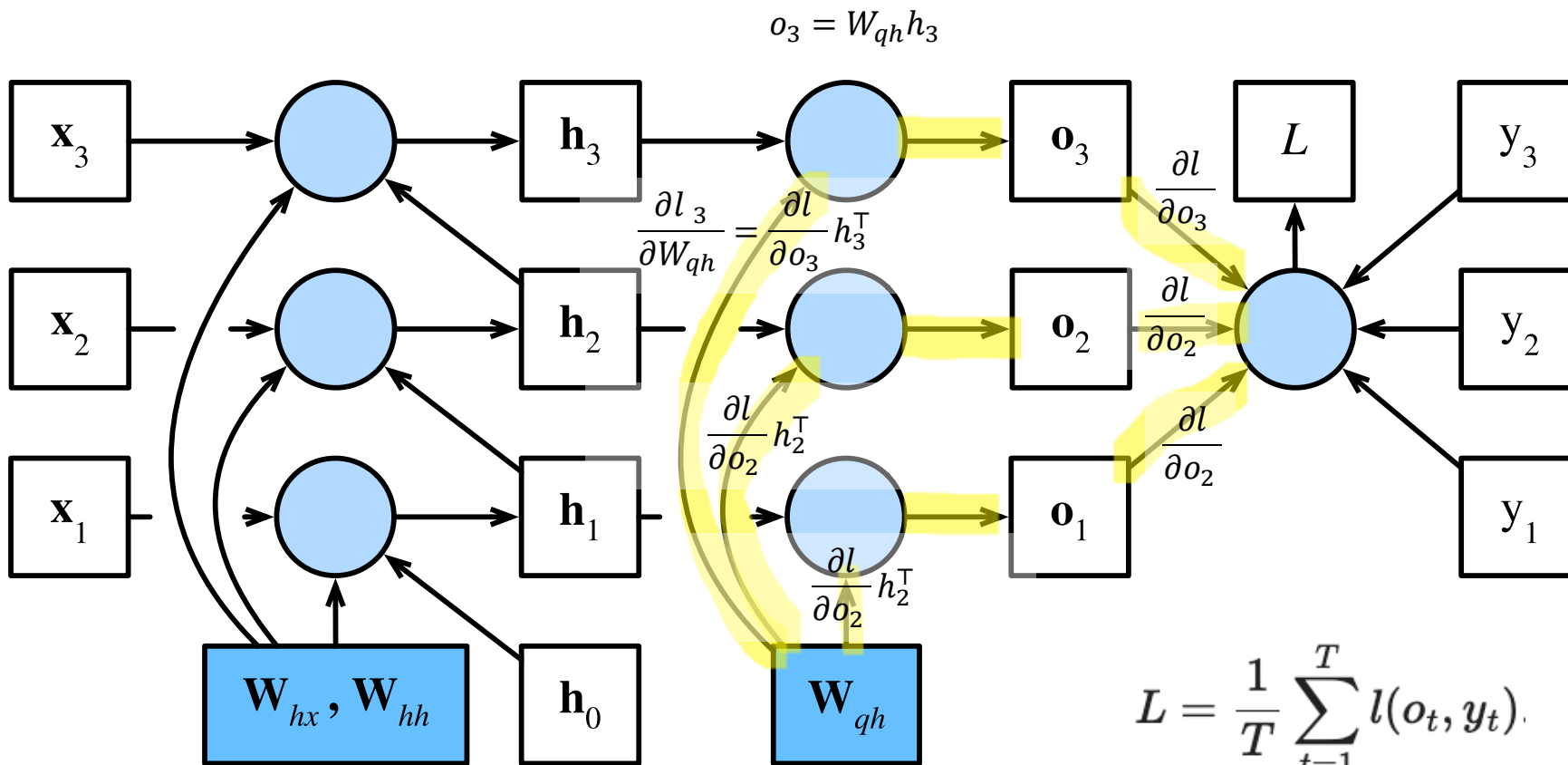
Step 1: $\frac{\partial L}{\partial W_{qh}}$

BPTT



Step 1: $\frac{\partial L}{\partial \mathbf{W}_{qh}}$

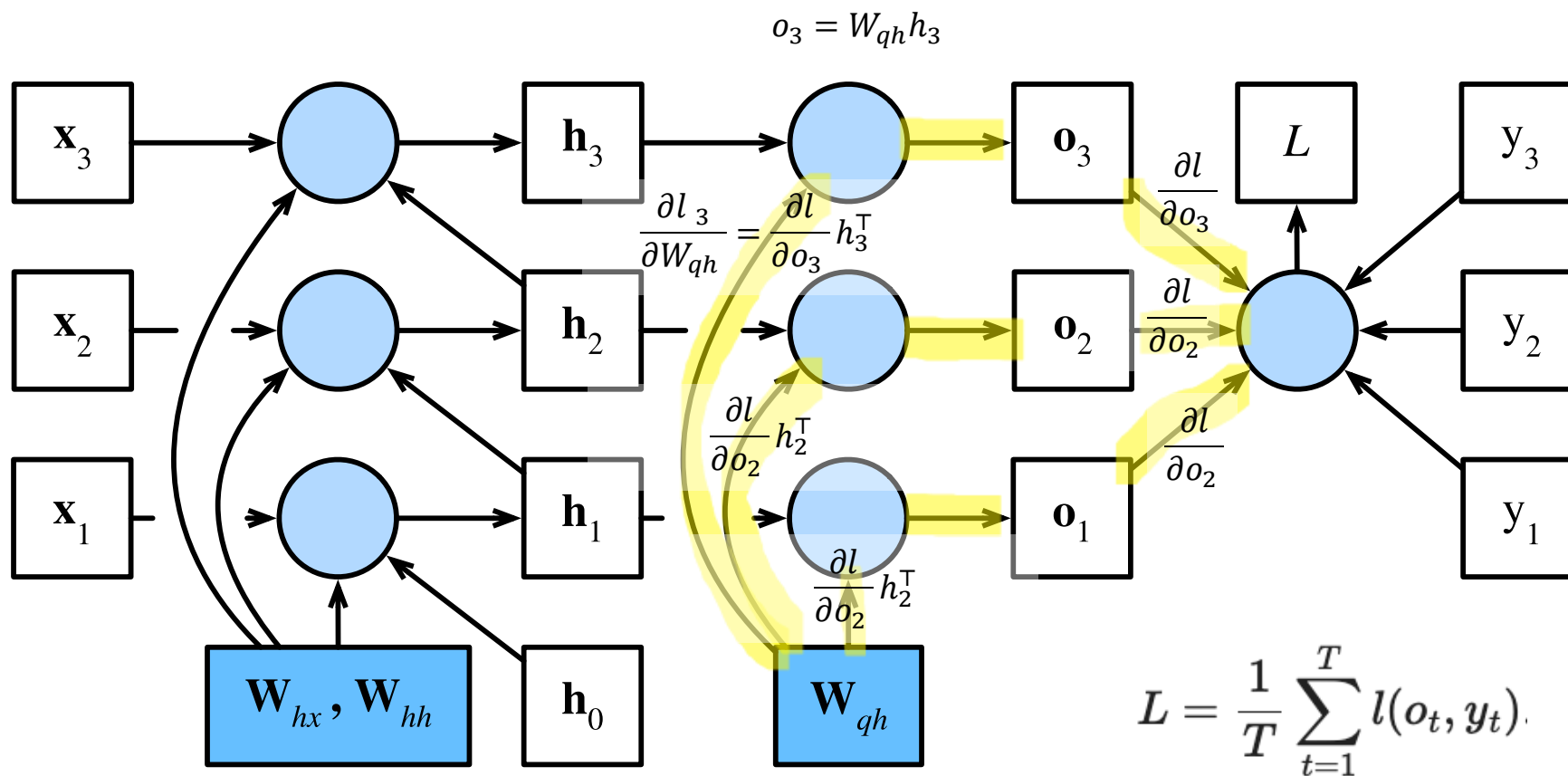
BPTT



$$L = \frac{1}{T} \sum_{t=1}^T l(o_t, y_t).$$

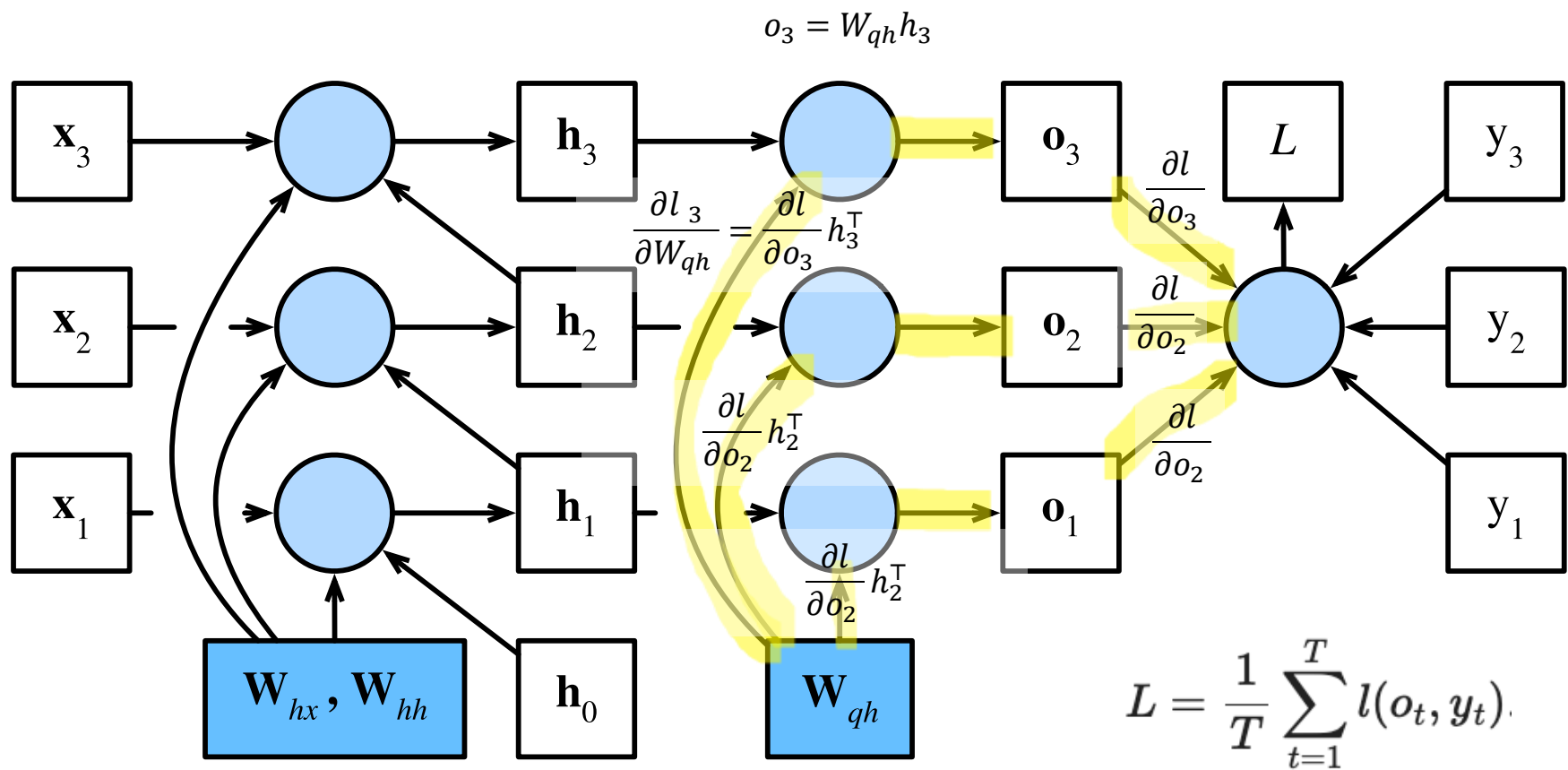
Step 1: $\frac{\partial L}{\partial W_{qh}}$

BPTT



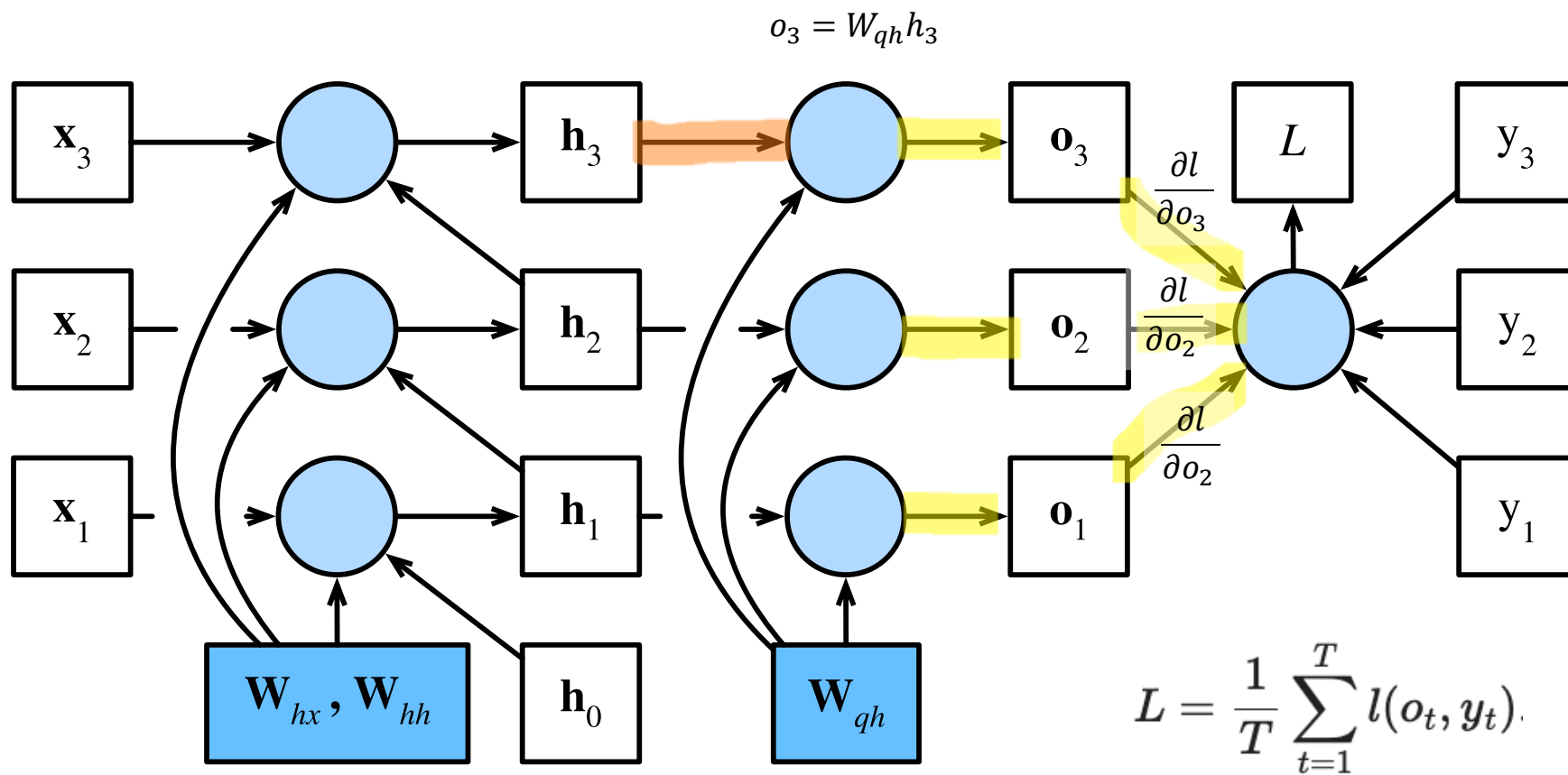
Step 1: $\frac{\partial L}{\partial W_{qh}} = \sum_{t=1}^T \frac{\partial l_t}{\partial W_{qh}} = \sum_{t=1}^T \frac{\partial l}{\partial t} h_t^\top$

BPTT



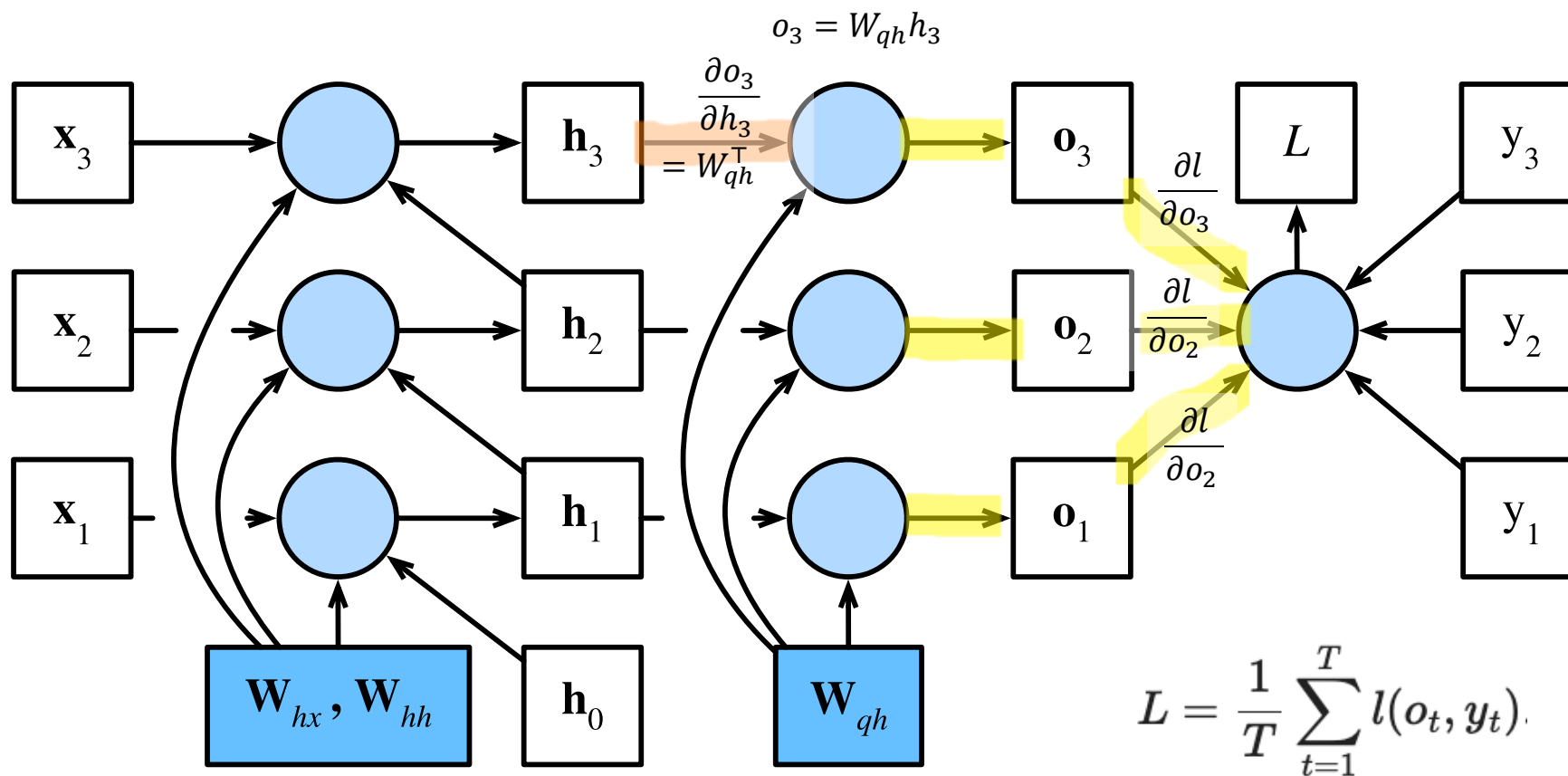
Next: $\frac{\partial L}{\partial W_{hh}}, \frac{\partial L}{\partial W_{hx}}$

BPTT



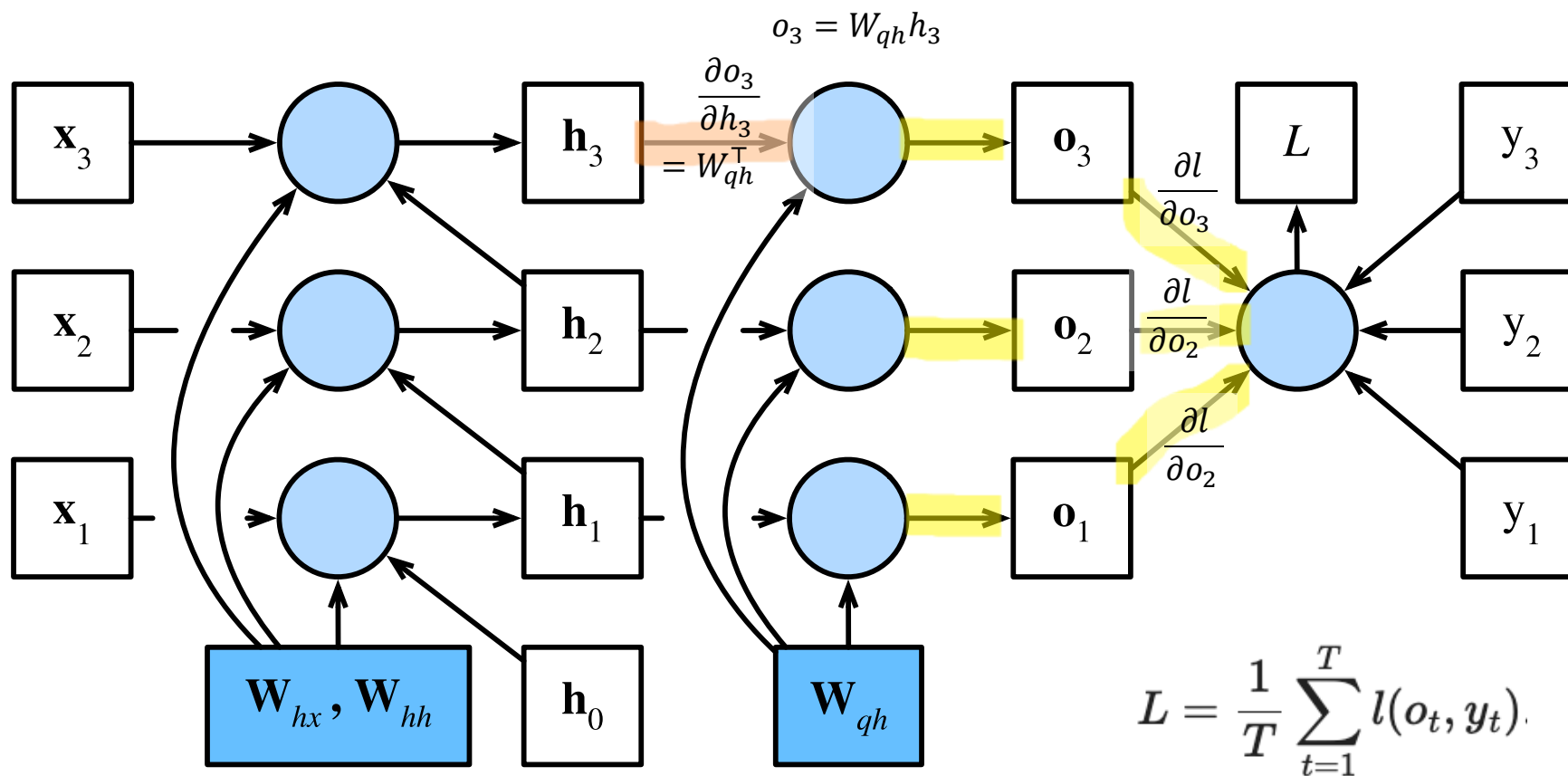
Step 2: $\frac{\partial L}{\partial h_T}$

BPTT



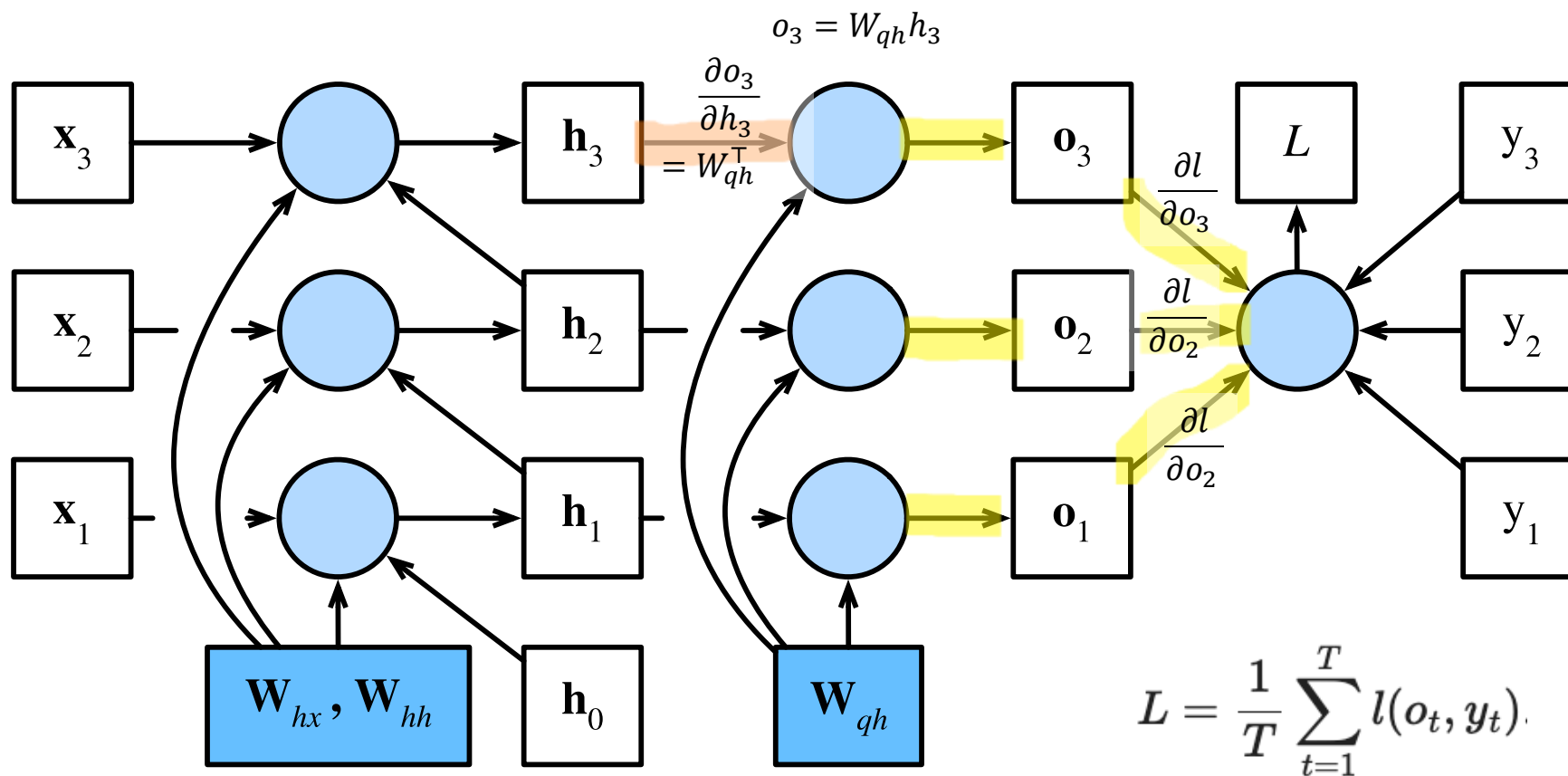
Step 2: $\frac{\partial L}{\partial \mathbf{h}_T}$

BPTT



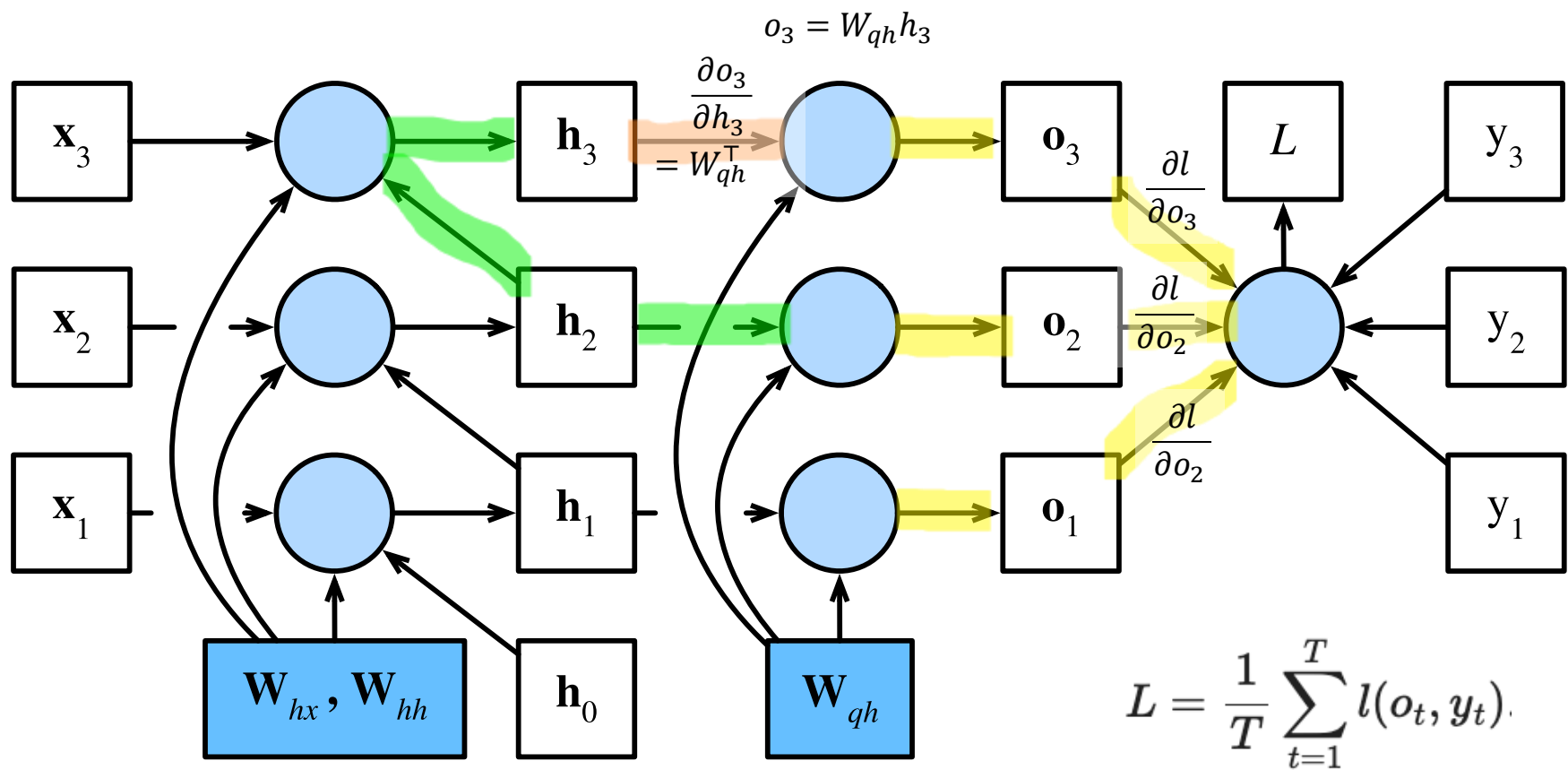
Step 2: $\frac{\partial L}{\partial h_T} = \frac{\partial l}{\partial o_T} \frac{\partial o_T}{\partial h_T} = W_{qh}^\top \frac{\partial l}{\partial o_T}$

BPTT



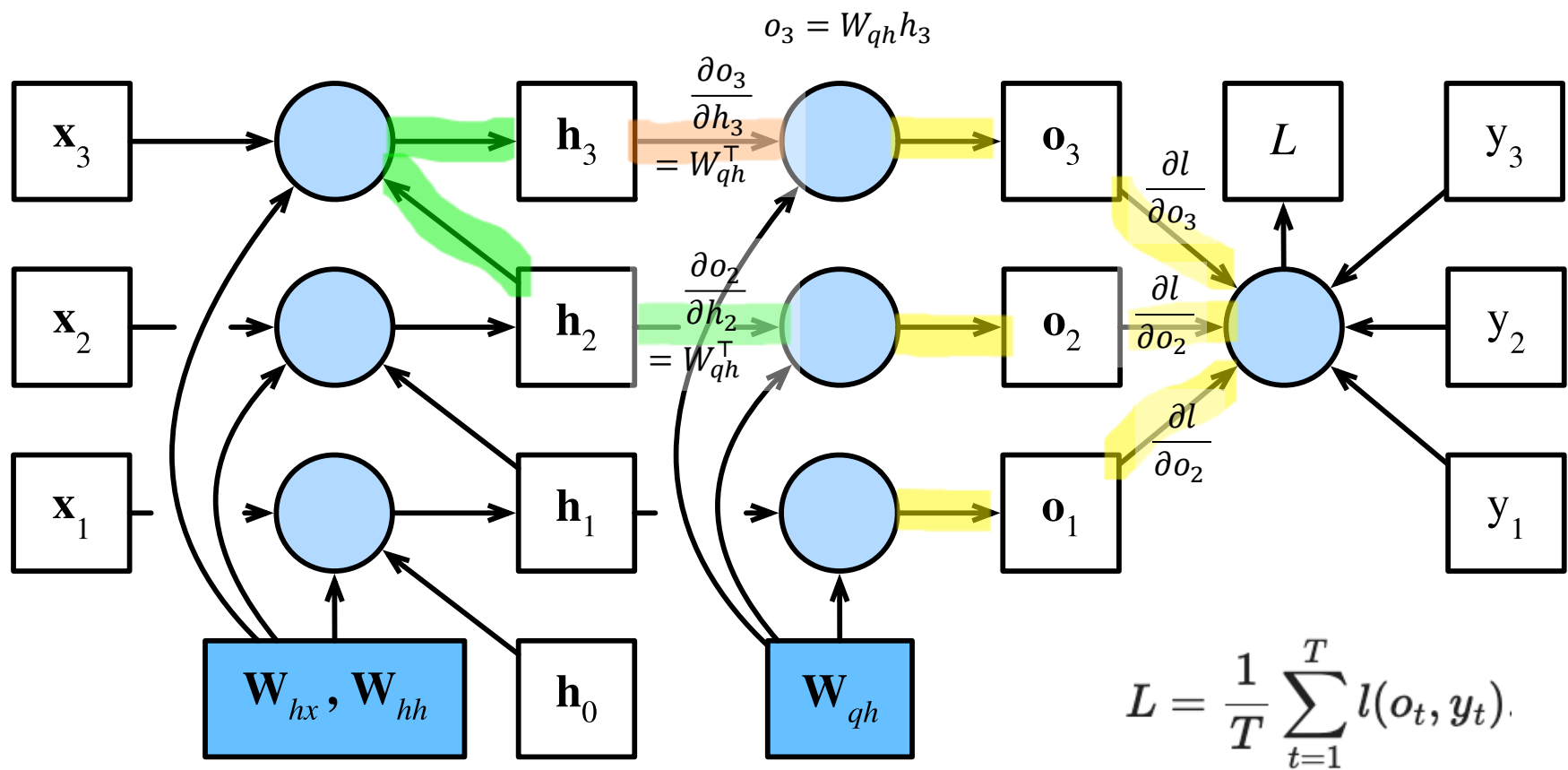
Step 3: $\frac{\partial L}{\partial \mathbf{h}_t}$

BPTT



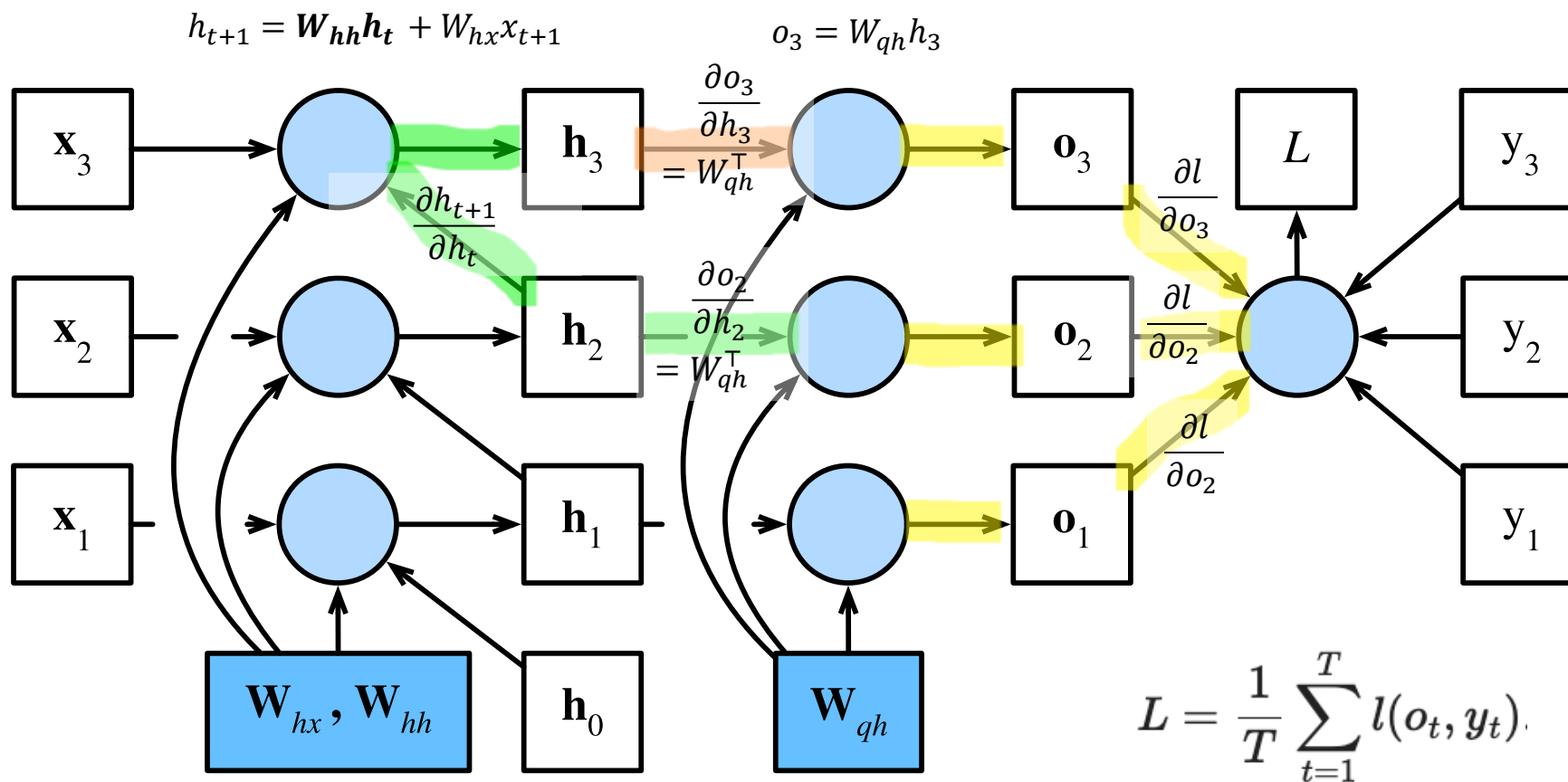
Step 3: $\frac{\partial L}{\partial \mathbf{h}_t}$

BPTT



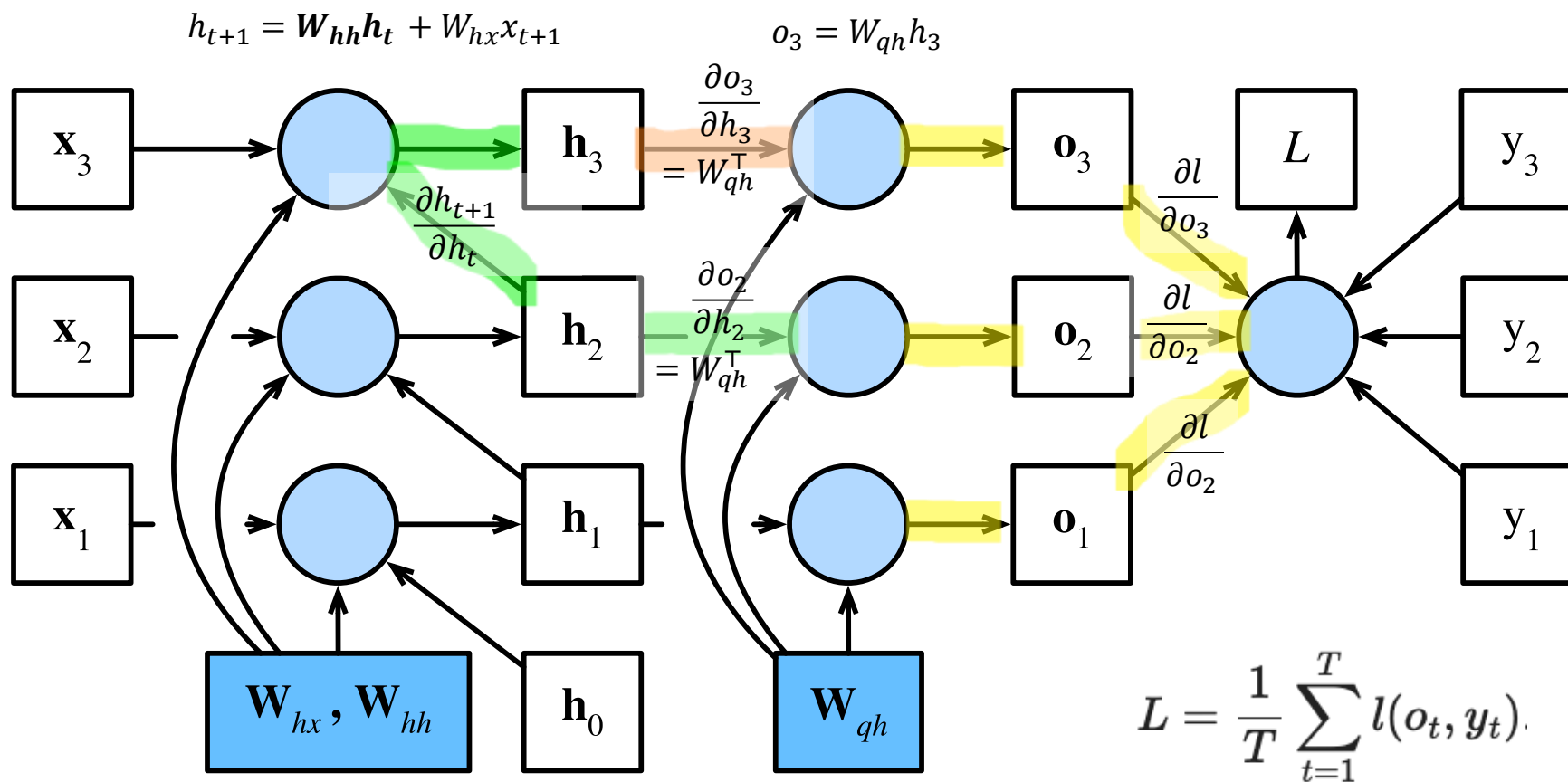
Step 3: $\frac{\partial L}{\partial \mathbf{h}_t}$

BPTT



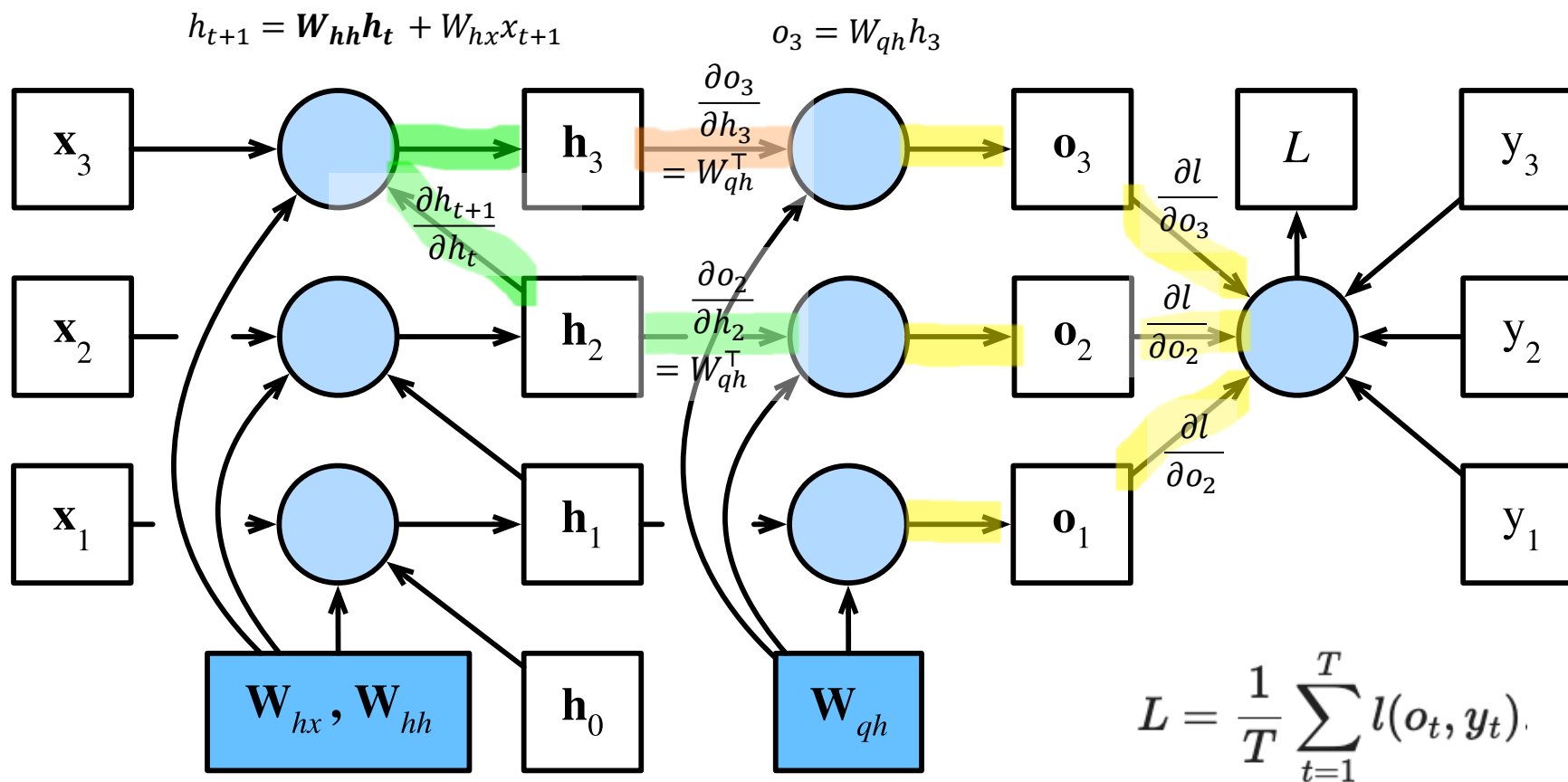
Step 3: $\frac{\partial L}{\partial \mathbf{h}_t}$

BPTT



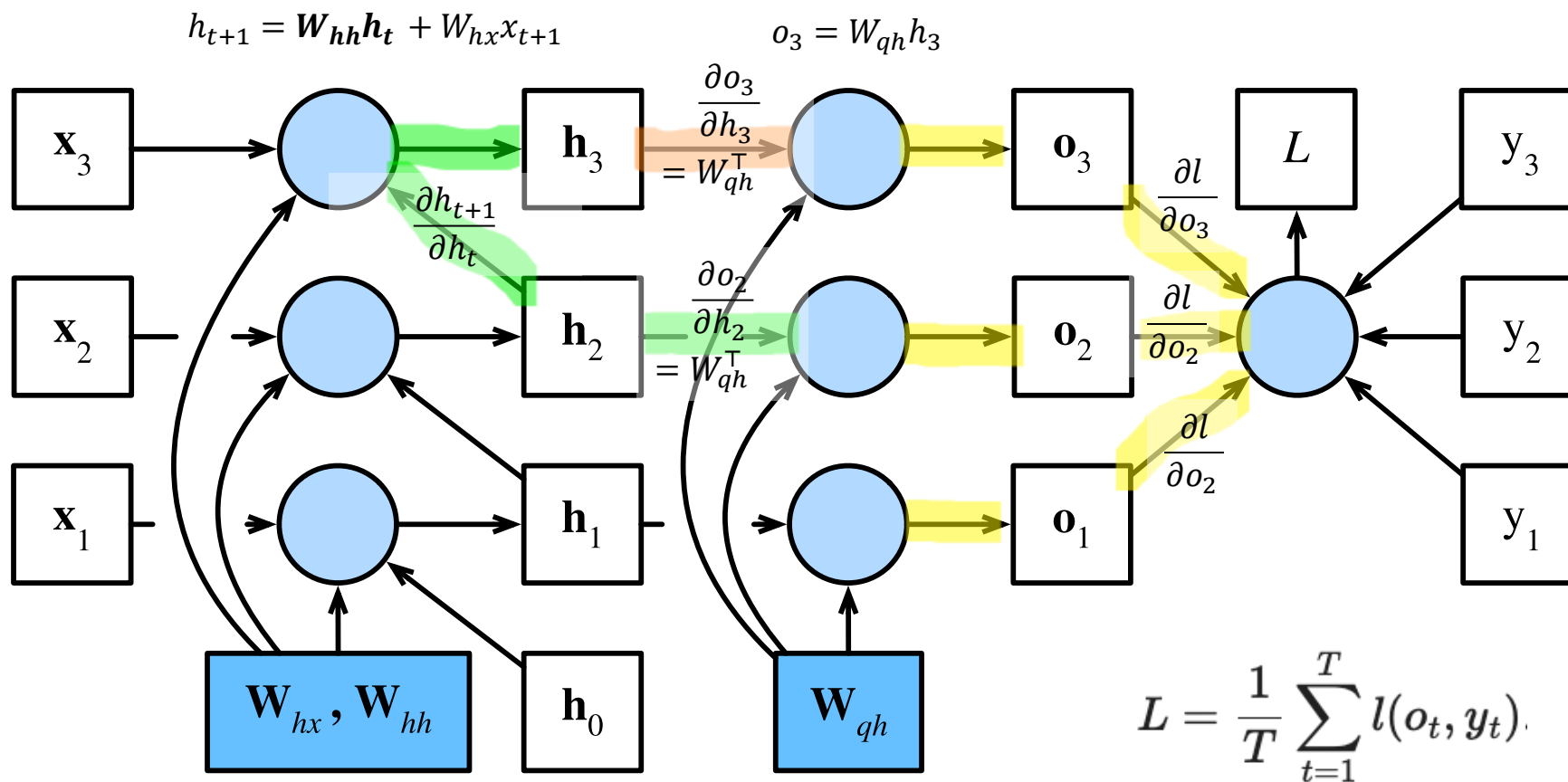
Step 3:
$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_t} + \frac{\partial L}{\partial o_t} \frac{\partial o_t}{\partial h_t}$$

BPTT



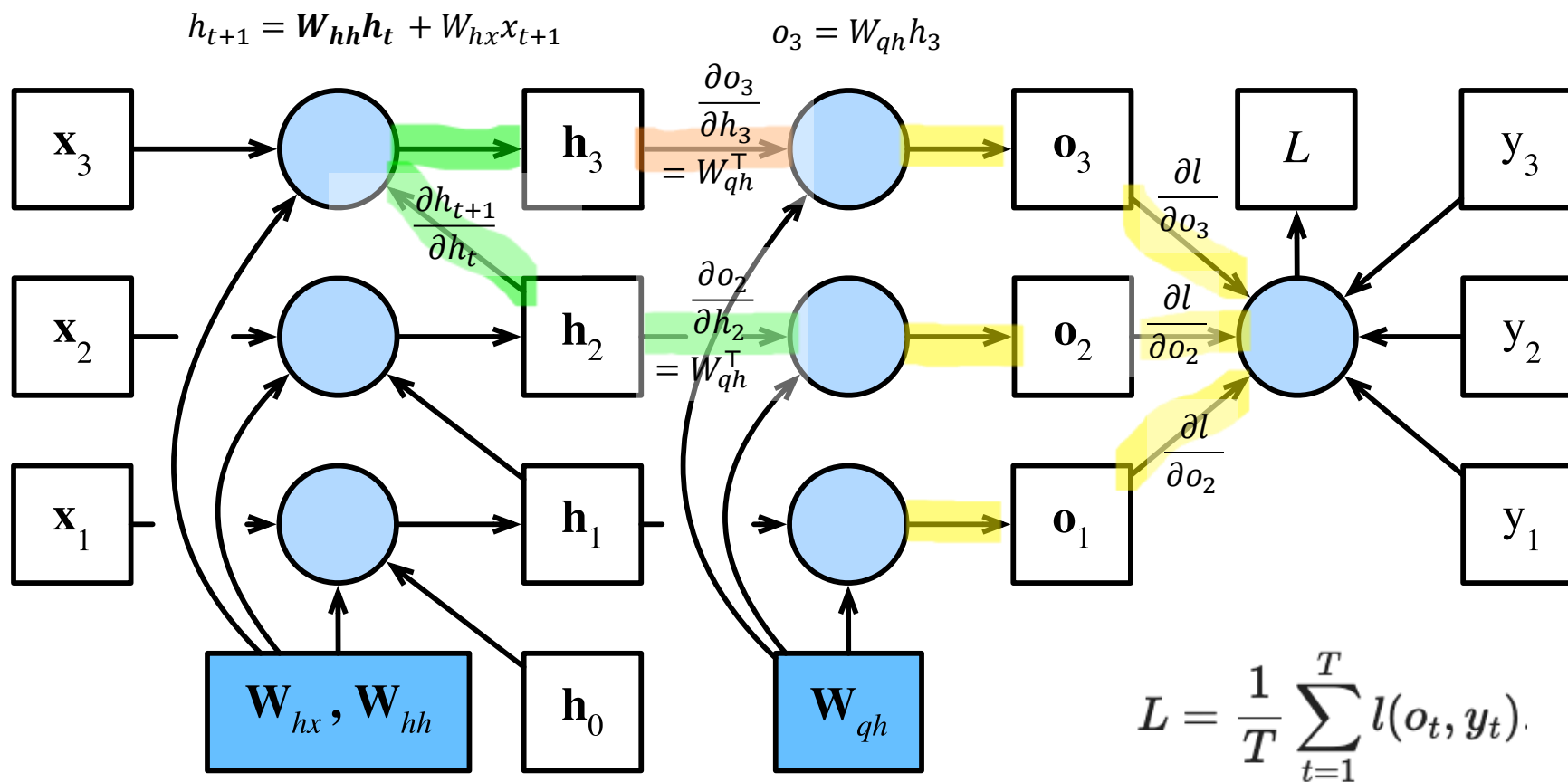
Step 3:
$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_t} + \frac{\partial L}{\partial o_t} \frac{\partial o_t}{\partial h_t} = W_{hh}^\top \frac{\partial L}{\partial h_{t+1}} + W_{qh}^\top \frac{\partial L}{\partial o_t}$$

BPTT



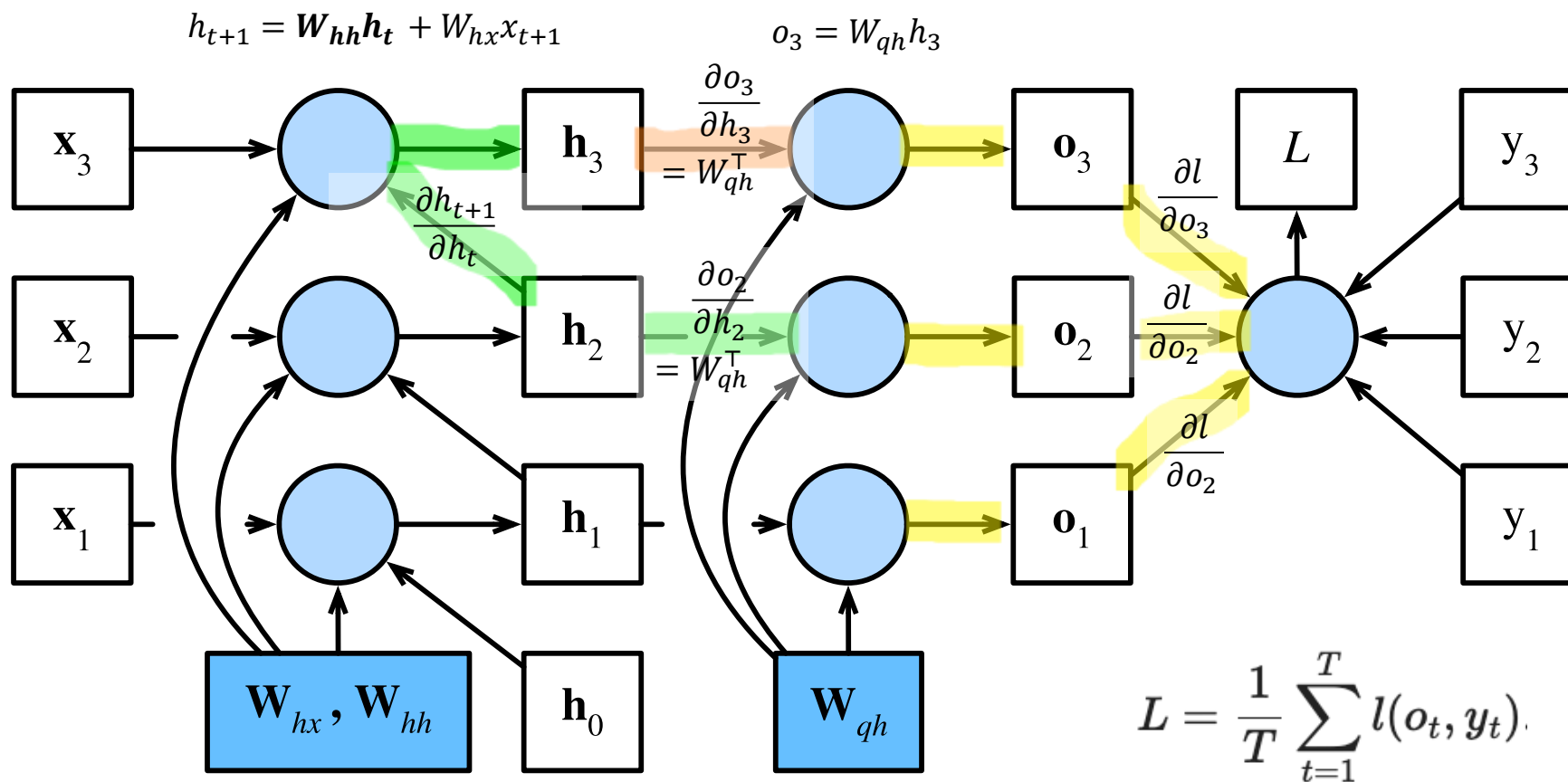
Step 3: $\frac{\partial L}{\partial h_t} = \mathbf{W}_{hh}^\top \frac{\partial L}{\partial h_{t+1}} + \mathbf{W}_{qh}^\top \frac{\partial L}{\partial o_t}$

BPTT



Step 3: $\frac{\partial L}{\partial \mathbf{h}_t} = \mathbf{W}_{hh}^\top \frac{\partial L}{\partial \mathbf{h}_{t+1}} + \mathbf{W}_{qh}^\top \frac{\partial L}{\partial \mathbf{o}_t}$ ← Recursive function

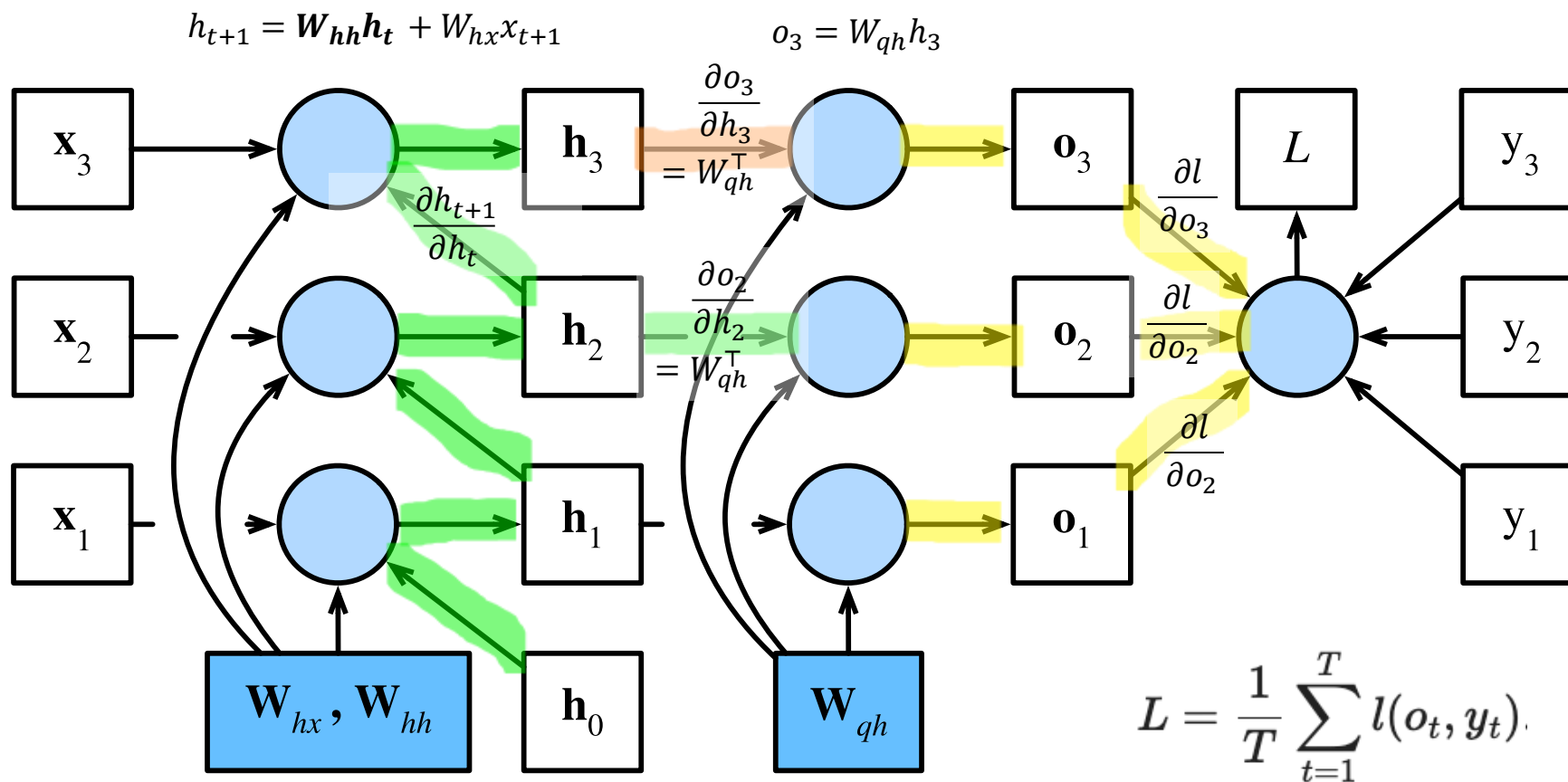
BPTT



Step 3: $\frac{\partial L}{\partial h_t} = W_{hh}^\top \frac{\partial L}{\partial h_{t+1}} + W_{qh}^\top \frac{\partial L}{\partial o_t}$ ← Recursive function

In general: $\frac{\partial L}{\partial h_t} = \sum_{i=t}^T (W_{hh}^\top)^{T-i} W_{qh}^\top \frac{\partial L}{\partial o_{T+i-i}}$

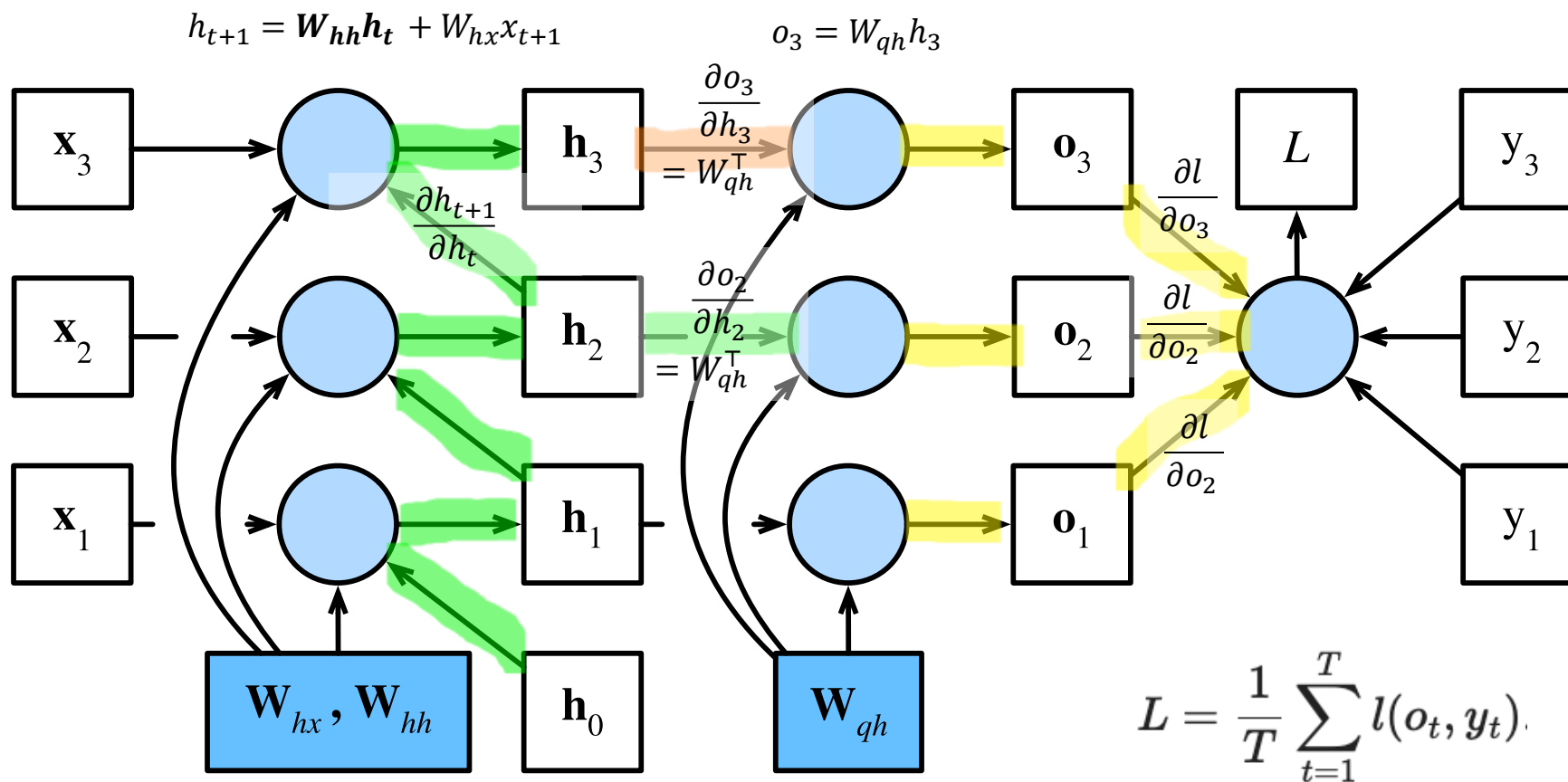
BPTT



Step 3: $\frac{\partial L}{\partial h_t} = W_{hh}^\top \frac{\partial L}{\partial h_{t+1}} + W_{qh}^\top \frac{\partial L}{\partial o_t}$ ← Recursive function

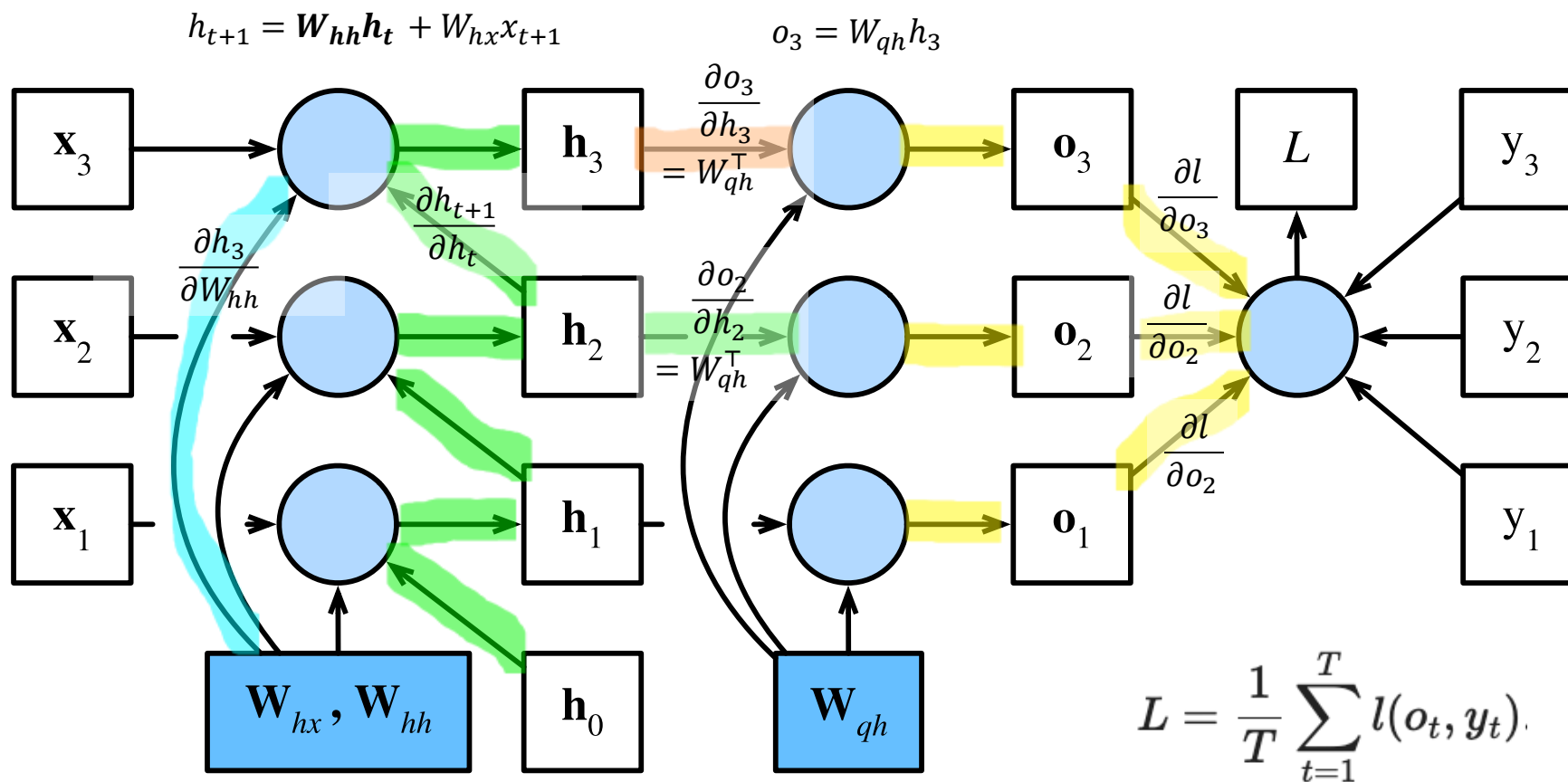
In general: $\frac{\partial L}{\partial h_t} = \sum_{i=t}^T (W_{hh}^\top)^{T-i} W_{qh}^\top \frac{\partial L}{\partial o_{T+i-i}}$

BPTT



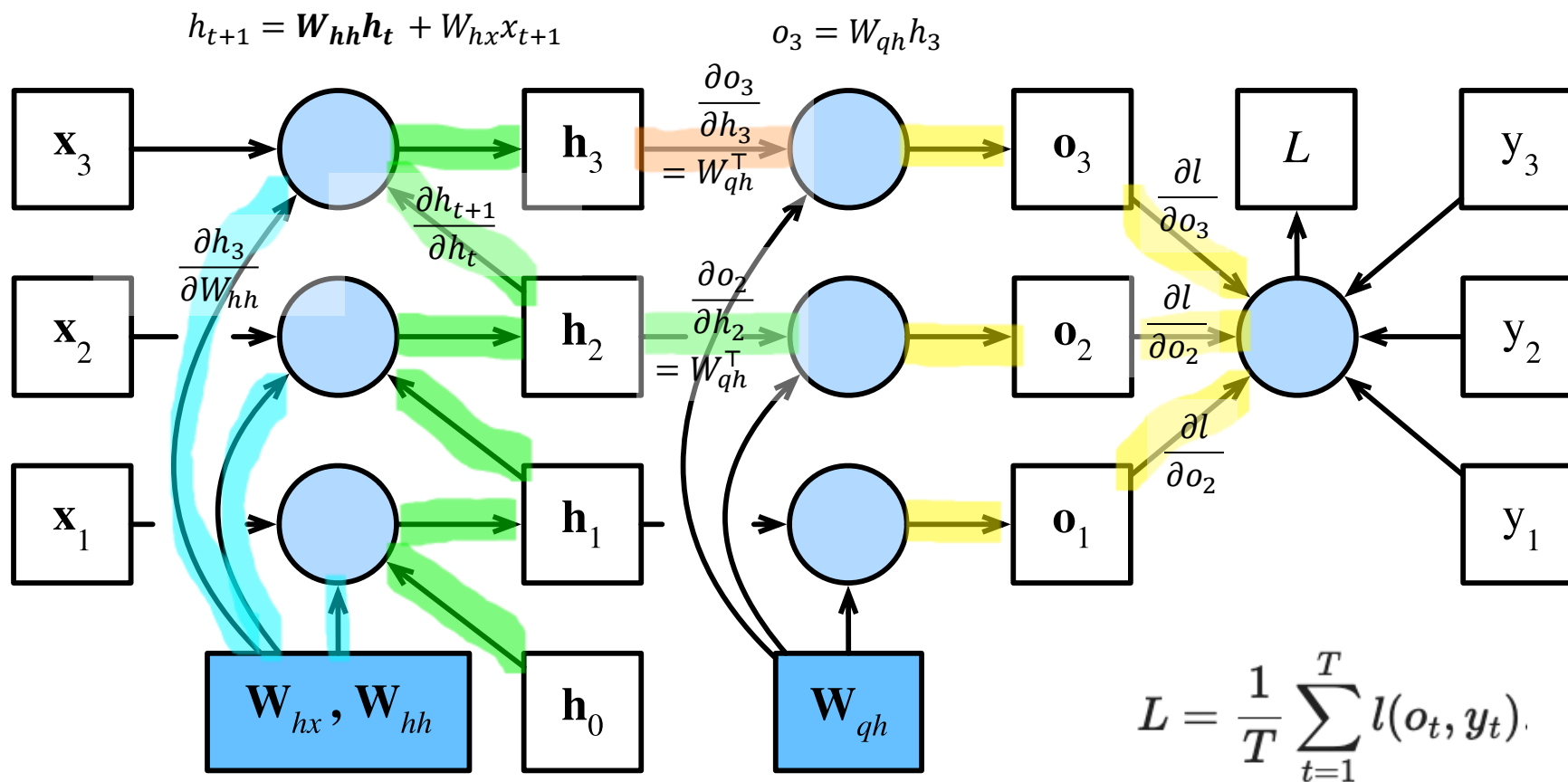
Step 4: $\frac{\partial L}{\partial \mathbf{W}_{hh}} =$

BPTT



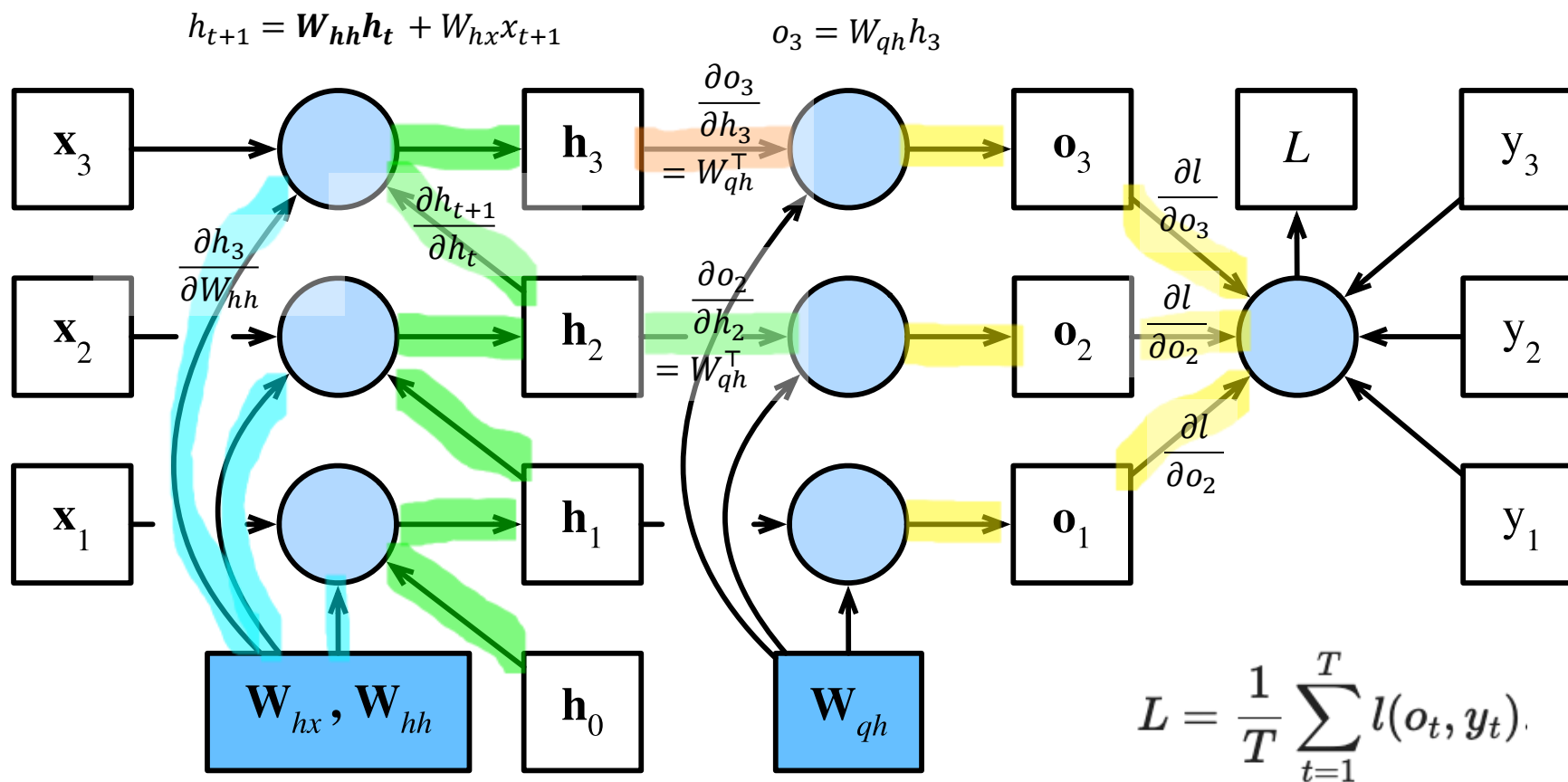
Step 4: $\frac{\partial L}{\partial \mathbf{W}_{hh}} =$

BPTT



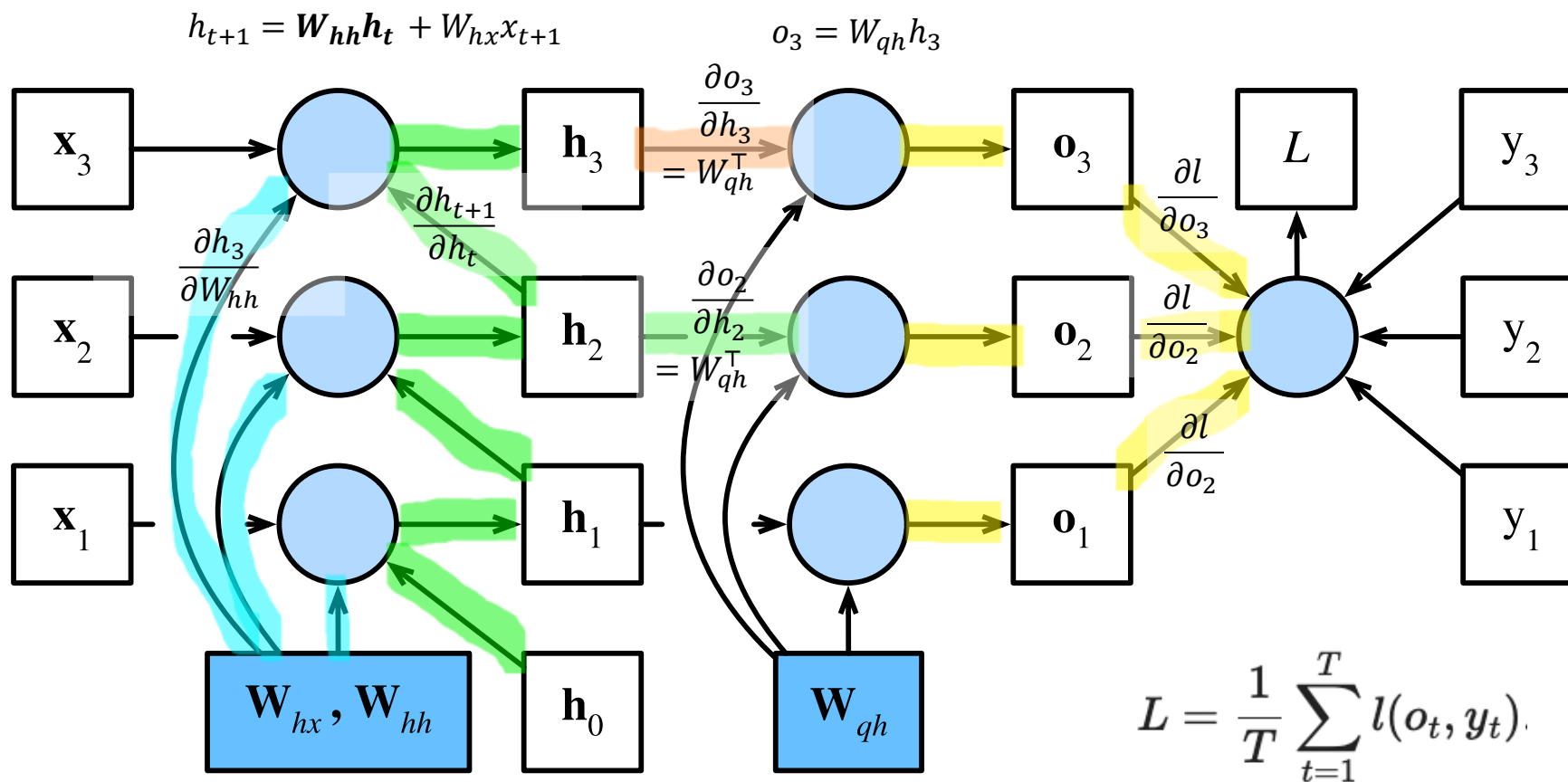
Step 4: $\frac{\partial L}{\partial \mathbf{W}_{hh}} =$

BPTT



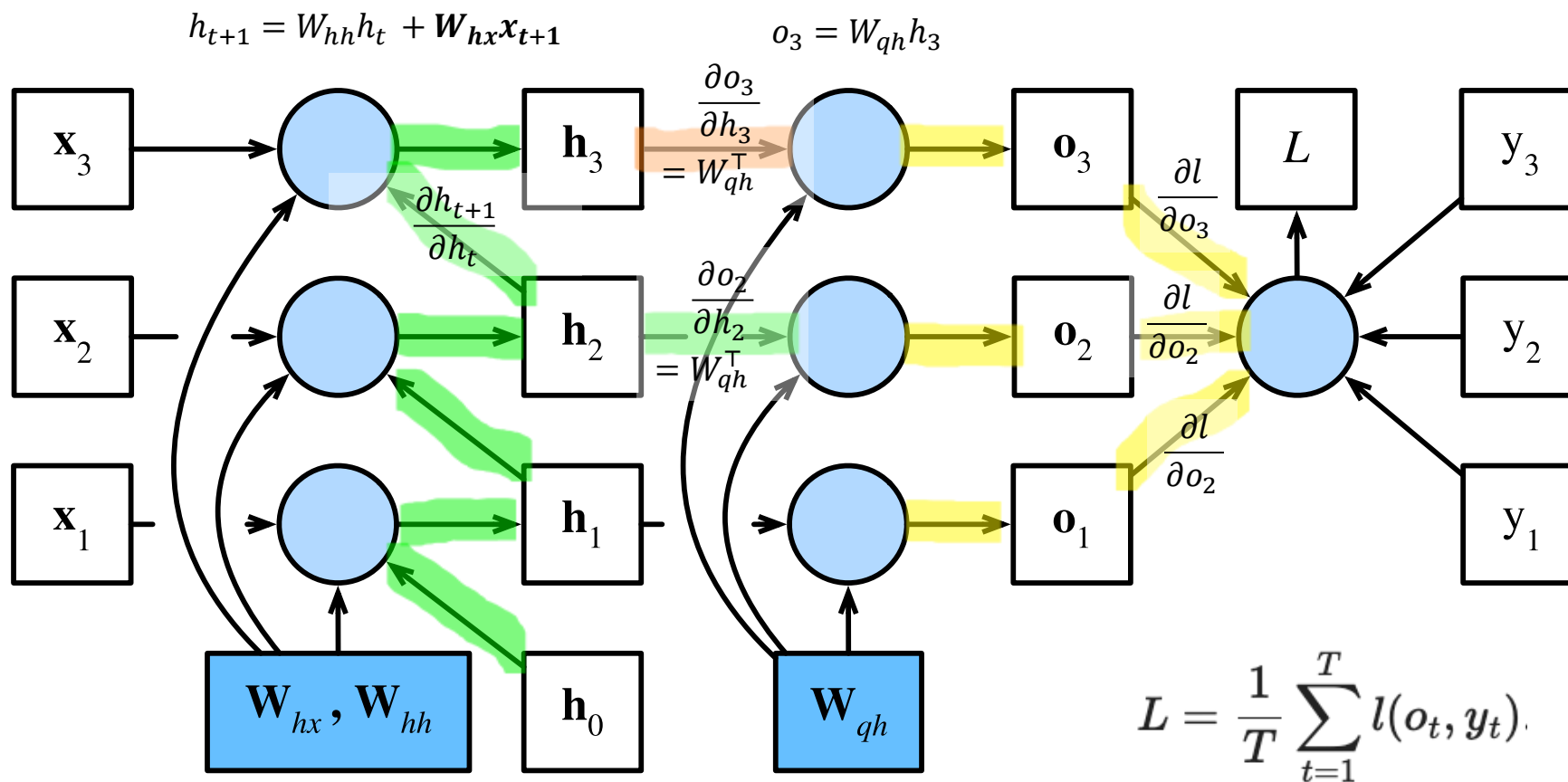
Step 4: $\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^T \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial W_{hh}} =$

BPTT



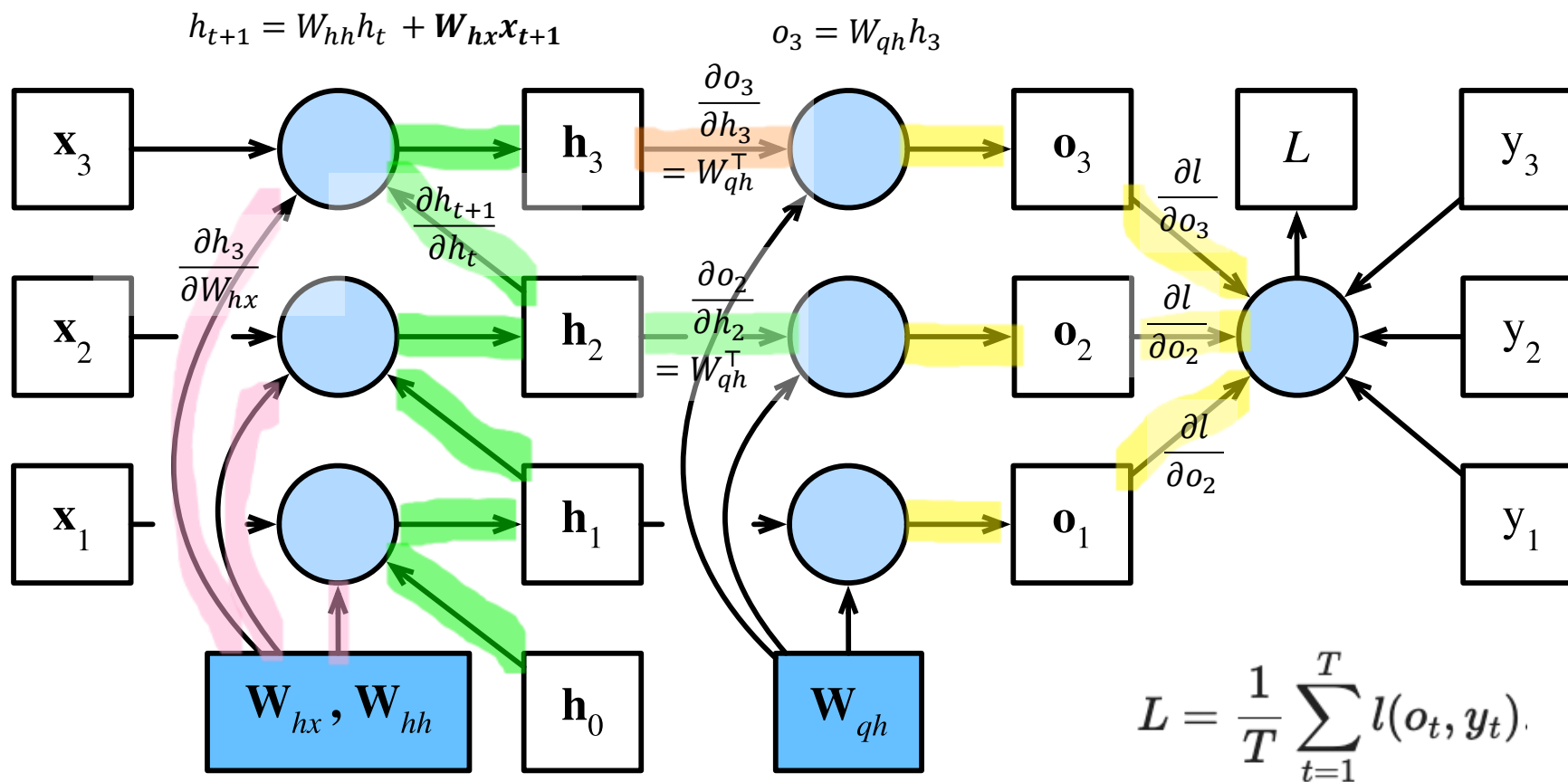
Step 4: $\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^T \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial W_{hh}} = \sum_{t=1}^T \frac{\partial L}{\partial h_t} h_{t-1}^\top$

BPTT



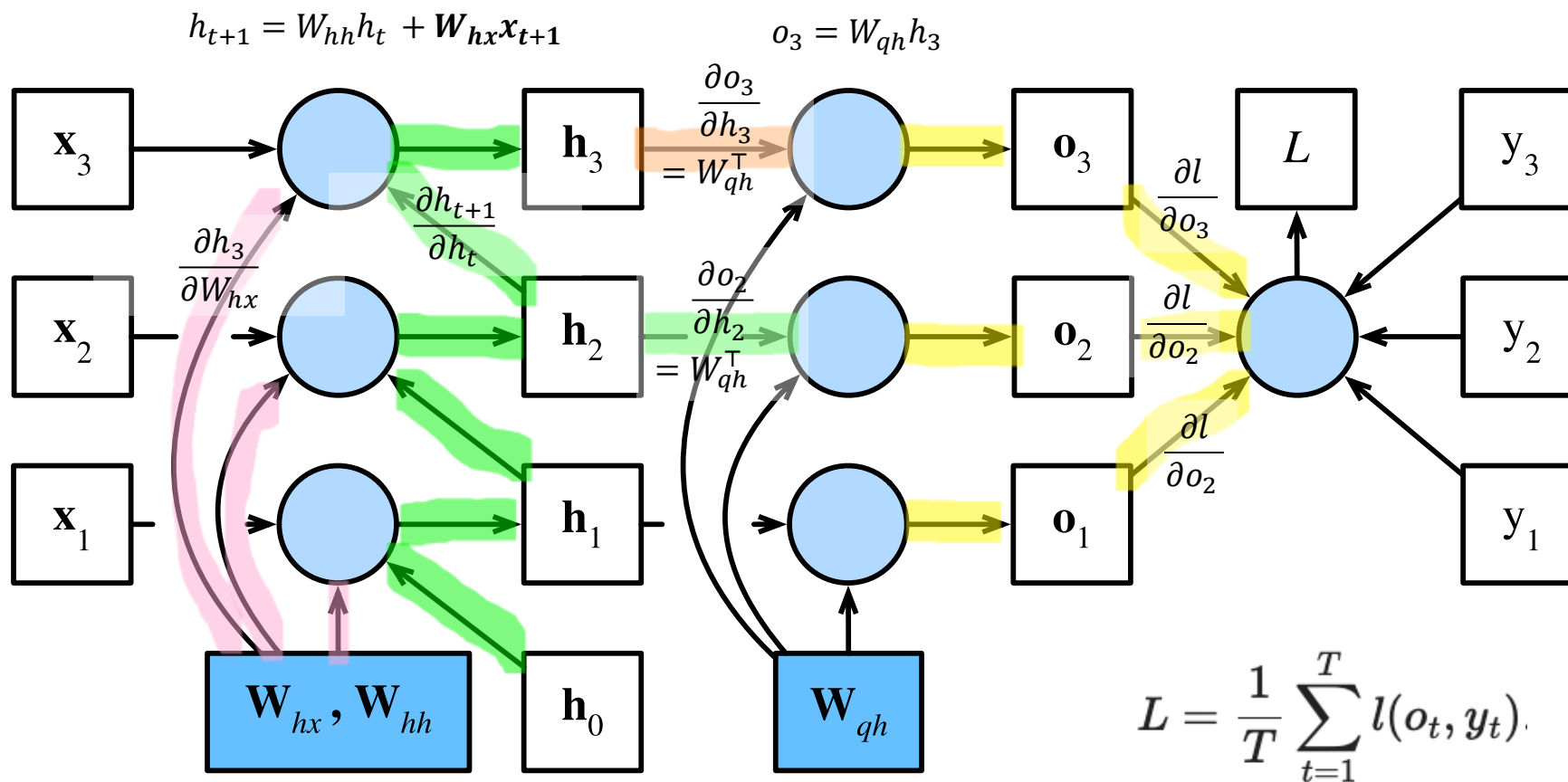
Step 4: $\frac{\partial L}{\partial \mathbf{W}_{hx}}$

BPTT



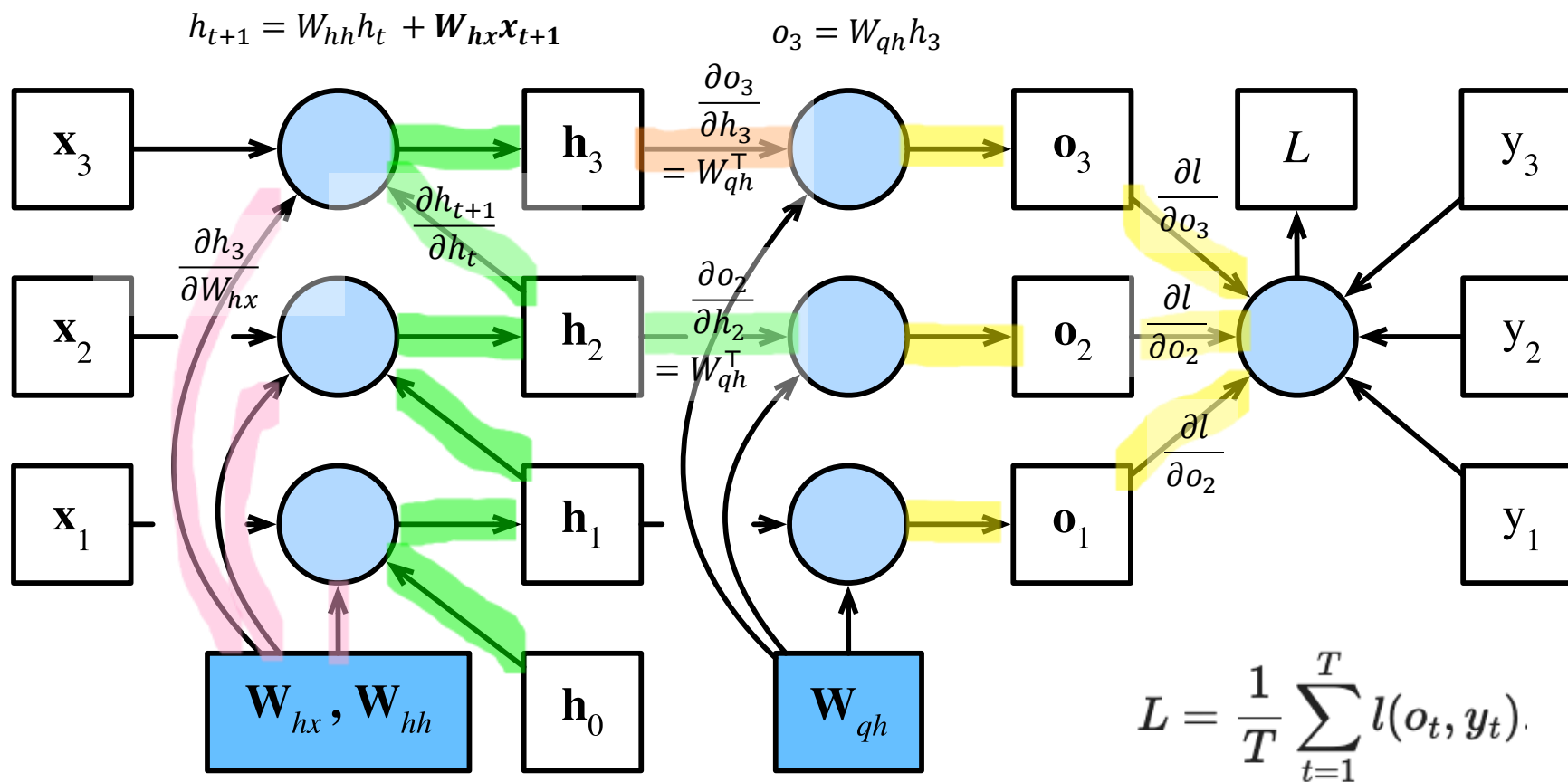
Step 4: $\frac{\partial L}{\partial W_{hx}}$

BPTT



Step 4: $\frac{\partial L}{\partial \mathbf{W}_{hx}} = \sum_{t=1}^T \frac{\partial L}{\partial \mathbf{h}_t} \frac{\partial \mathbf{h}_t}{\partial \mathbf{W}_{hx}} =$

BPTT



Step 4: $\frac{\partial L}{\partial W_{hx}} = \sum_{t=1}^T \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial W_{hx}} = \sum_{t=1}^T \frac{\partial L}{\partial h_t} x_t^\top =$